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Inventor(s) Ryder; Noah Lael et al.

Fire mitigation system for energy storage systems

Abstract

A system for mitigating fire within a battery storage container enclosing an energy storage system includes a sensor configured to detect a precursor condition indicative of a potential fire or explosion, a controller, and a set of extendable battery trays, each including a tray ejector and containing a set of battery cells. The controller detects a precursor condition in a battery tray via the sensor and ejects the battery tray to increase the distance between the battery tray and adjacent battery trays. The system can include a cooling channel in the battery tray configured to cool the set of battery cells, and/or a nozzle configured to direct fluid into the battery tray to suppress the precursor condition. In one variation, the system includes a door of the container configured to open, venting the interior of the container in response to detection of a precursor condition indicating a potential explosion.

Inventors: **Ryder; Noah Lael (Rockville, MD), Turner, III; Brent Allen (Rockville, MD), Hutchens; Ryan Alexander (Rockville, MD), Natale; Anthony James (Rockville, MD)**

Applicant: **Fire & Risk Alliance, LLC** (Rockville, MD)

Family ID: **84490486**

Assignee: **Fire & Risk Alliance, LLC (Derwood, MD)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2021/0249639	12/2020	Shao	N/A	H01M 10/486

FOREIGN PATENT DOCUMENTS

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WO-2021136875	12/2020	WO	A62C 3/16

Primary Examiner: Lynch; Victoria H

Attorney, Agent or Firm: Run8 Patent Group, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application claims priority to U.S. Provisional Application No. 63/212,240, filed on 18 Jun. 2021, and to U.S. Provisional Application No. 63/299,792, filed on 14 Jan. 2022, both of which are incorporated in their entireties by this reference.

TECHNICAL FIELD
(1) This invention relates generally to the field of energy storage systems and more specifically to a

new and useful system for mitigating and preventing the spread of fires within an energy storage system.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. **1** is a schematic representation of one variation of a system;

(2) FIG. **2** is a flowchart representation of a method;

(3) FIG. **3** is a schematic representation of a variation of the system;

(4) FIG. **4** is a flowchart representation of the method;

(5) FIG. **5** is a flowchart representation of one variation of the method;

(6) FIG. **6** is a flowchart representation of one variation of the method;

(7) FIG. **7** is a flowchart representation of one variation of the method; and

(8) FIG. **8** is a flowchart representation of one variation of the method.

DESCRIPTION OF THE EMBODIMENTS

(9) The following description of embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention. Variations, configurations, implementations, example implementations, and examples described herein are optional and are not exclusive to the variations, configurations, implementations, example implementations, and examples they describe. The invention described herein can include any and all permutations of these variations, configurations, implementations, example implementations, and examples.

(10) 1. System

(11) As shown in FIGS. **1-8**, a system **100** for mitigating a fire within a battery storage container **110** enclosing an energy storage system including a set of battery cells **122**. As used herein, a cell or battery cell **122** can include an individual unit of energy storage capacity (e.g., battery, capacitor, etc.), a battery tray **120** can include multiple battery cells **122** arranged serially or in parallel within a battery storage container **110**, and a unit can include multiple battery trays **120** assembled together as a complete battery rack **111**. The system **100** can include: a fluid storage tank for storing a volume of fluid; the battery storage container **110**, a securing mechanism configured to secure the battery storage container **110** enclosing the set of battery cells **122**; a cooling channel **151** positioned within the battery storage container **110** and having a lumen disposed between a proximal end, a distal end of the cooling channel **151**, the proximal end connected to the fluid storage tank, and a set of apertures **152** contiguous with the cooling channel **151** and configured to eject the volume of fluid from the cooling channel **151**, and set of meltable plugs **153** arranged over the set of apertures **152** and configured to melt and expose the aperture **152** in response to an ambient temperature surrounding the system **100** exceeding a threshold temperature, thereby releasing the volume of fluid.

(12) In one variation, the system **100** can include: a battery rack **111**; a sensor **113** configured to detect a precursor condition **102** to an incipient fire event in the battery rack **111**; and a battery tray **120** configured to retain a set of battery cells **122**, occupy the battery rack **111** in an inserted position, and extend out of and be supported by the battery rack **111** in an extended position. The system **100** can further include: a first tray ejector **124** configured to transition the battery tray **120** from the inserted position to the extended position in response to detection of the precursor condition **102**. The system **100** can further include an intercooler **150** arranged in the battery tray **120** including a cooling channel **151** configured to circulate fluid to cool the set of battery cells **122** occupying the battery tray **120**; a supply manifold **157** arranged proximal the battery rack **111**, fluidly coupled to the intercooler **150**, and configured to supply fluid to the intercooler **150**; and a return manifold **158** arranged proximal the battery rack **111**, fluidly coupled to the intercooler **150**,

and configured to receive fluid from the intercooler **150**. The system **100** can further include a nozzle **114**: fluidly coupled to the supply manifold **157**; arranged in the battery tray **120**; and configured to receive fluid from the supply manifold **157** and direct fluid into the battery tray **120** in response to detection of the precursor condition **102**.

(13) In another variation, the system **100** can include: a nozzle **114** including an inlet connected to the cooling channel **151** and an outlet positioned to direct a fluid spray pattern within the battery storage container **110** enclosing the set of battery cells **122**; and a meltable plug **153** arranged over the outlet of the nozzle **114** and configured to melt and expose the outlet in response to an ambient temperature surrounding the system **100** exceeding a threshold temperature.

(14) In another variation, the system **100** includes a flow detection sensor configured to detect a flow of fluid from the fluid storage tank, into the cooling channel **151**, and out of the set of apertures **152** of the cooling channel **151** or the outlet of the nozzle **114**; and a controller **130** configured to, in response to the flow detection sensor detecting the flow of fluid through the outlet of the nozzle **114**, transmit a warning prompt to a remote monitoring system.

(15) In another variation, the system **100** also includes: a pump **155** connected to the fluid storage tank and configured to draw fluid out of the fluid storage tank, into the cooling channel **151**, and out of the outlet of the nozzle **114**; and a sensor **113** (e.g., a temperature, humidity, gas/vapor, and/or light sensor **113**) configured to detect a change in ambient conditions within the battery storage container **110**. In this variation, the system **100** can also include: a directional control valve **156** configured to open and release fluid to flow from the fluid storage tank, through the cooling channel **151**, and through the nozzle **114** directing the fluid at the set of battery cells **122** within the battery storage container **110**; and a controller **130** configured to, in response to the sensor **113** detecting the change in ambient conditions within the battery storage container **110** exceeding a threshold (e.g., a change in temperature exceeding 100 degrees Celsius), activate the directional control valve **156** to open and activate the pump **155** to draw fluid out of the fluid storage tank, into the cooling channel **151**, and out of the outlet of the nozzle **114**.

(16) In another variation, the system **100** includes a waste tank **160** (e.g., storage volume) arranged in the bottom of the battery storage container no and configured to collect fluid deposited within the battery storage container **110**—responsive to a fire event —to prevent this fluid from escaping the battery storage container **110** and contaminating an environment external to the battery storage container **110**.

(17) In another variation, the system **100** includes a door **115** arranged within the perimeter wall of the battery storage container **110** and configured to open in response to the controller **130** detecting flammable and/or explosive gasses within the battery storage container **110** in order to prevent buildup of flammable and/or explosive gasses within the battery storage container no and thus reducing risk of explosion within the battery storage container **110**.

(18) In another variation, the system **100** includes a battery tray **120**: configured to contain a set of battery cells **122**; and configured to automatically eject from (e.g., extend laterally out from) a battery rack in in response to a sensor **113** (or the controller **130**) detecting a precursor condition **102** or a fire event within the battery tray **120** in order to reduce risk of the fire event propagating to an adjacent battery tray **120** within the battery storage container no.

(19) 1.1 Method

(20) As shown in FIG. **8**, the system **100** can execute blocks of a method S**100** to detect and mitigate a fire within a battery storage container including: at a battery tray **120** arranged within a battery rack in, retaining a set of battery cells **122** within the battery tray **120** in Block S**108**; and circulating fluid through an intercooler **150** arranged within the battery tray **120** to cool the set of battery cells **122** occupying the battery tray **120** in Block S**130**. The method S**100** can further include, at a directional control valve **156** fluidly coupled to a supply manifold **157**, the intercooler **150**, and a nozzle **114** arranged within the battery tray **120**: receiving fluid from the supply manifold **157**; and supplying fluid to the intercooler **150** in Block S**142**. The method S**100** can

further include: at a sensor **113** arranged within the battery tray **120**, detecting a precursor condition **102** to an incipient fire event in the first battery tray **120** in Block Silo; in response to detection of the precursor condition **102** by the sensor **113**, extending the battery tray **120** from the battery rack **111** in Block S**120**; and, in response to detection of the precursor condition **102** by the sensor **113**, controlling the directional control valve **156** to transition from a first state supplying fluid to the intercooler **150**, to a second state supplying the fluid to the nozzle **114** in Block S**140**. The method S**100** can further include, at a controller **130**: receiving a signal from the sensor **113** indicating detection of the precursor condition **102** to an incipient fire event in Block S**160**; in response to receiving the signal from the sensor **113**, generating a notification indicating a fire event present in the battery tray **120** in Block S**164**; and transmitting the notification to an operator in Block S**166**. In one variation, the method S**100** can further include electrically isolating the battery tray **120** exhibiting the precursor condition **102** in Block S**150**. In another variation, the method S**100** can further include, at a controller **130**, calculating a risk value of a human approaching the battery tray **120** in Block S**162**.

(21) 2. Applications

(22) Generally, the system **100** is configured to: rapidly respond to an incidence of an ongoing fire event within a battery storage container **110** due to combustion of lithium-ion battery cells stored within the battery storage container **110**; suppress the fire event; and mitigate the risks of heat propagation and secondary fire events due to the initial fire event. In particular, the system **100** can direct a fire suppression fluid (e.g., water, inert gas, fire suppression agent, or some combination thereof) to: suppress the ongoing fire event at a particular enflamed lithium battery cell; minimize the propagation of the fire to adjacent lithium battery cells within the battery storage container **110**; and minimize the propagation of the fire to adjacent battery trays **120** storing additional lithium battery cells, or to adjacent battery storage containers **110**. Furthermore, the system **100** can direct the fire suppression fluid to decrease the ambient temperature within the battery storage container no and minimize the risks of “thermal runaway” (i.e., when elevated temperatures accelerate an energy release by a lithium battery cell that further increases temperatures and can have cascading adverse effects on nearby lithium battery cells), which can otherwise produce an explosive environment within the battery storage container no and increase the probability of a secondary fire and/or explosion.

(23) By decreasing temperatures within adjacent battery trays **120**, the system minimizes the risk of thermal runaway propagation outside the battery tray **120** of origin by diminishing the ability of adjacent battery trays **120** to potential combustible gases or other thermal injury. By reducing the overall number of cells involved in a failure event, the overall quantity of combustible gases is also reduced, thereby reducing the overall thermal exposure to the adjacent cells. As described below in more detail, the system: suppresses any potential fires via the fire suppression fluid while simultaneously ventilating the battery trays **120** via ventilation systems so that the fire can be extinguished while the battery tray(s) **120** is being ventilated.

(24) In one implementation, in addition to decreasing temperatures within adjacent battery trays **120**, the system is configured to maximize the distance between adjacent battery trays **120** in response to a first battery tray **120** exhibiting a fire event or precursor condition **102** to a fire event. In response to detection of the precursor condition **102**, the first battery tray **120** is ejected from the battery rack **111**, maximizing the distance between the first battery tray **120** and a second adjacent battery tray **120**. Increasing the distance between the first battery tray **120** and the second battery tray **120** reduces the potential for heat transfer between the battery trays **120**, thereby interrupting a potential chain reaction of heat propagation through the set of battery cells **122** within the battery storage container **110**, potentially causing thermal runaway in a multitude of battery cells **122** and/or destruction of the entire battery storage container **110**.

(25) In another implementation, the system **100** is an enclosed apparatus—with a fluid storage tank storing the fire suppression fluid—that can be installed and secured within or external to various

types of battery storage containers **110**. The system **100** can aerosolize the fire suppression fluid such that the fluid behaves like a gas (i.e., suspended in air) and moves like a gas throughout the battery storage container **110** to: deposit on vertical, horizontal, and angled surfaces and in between the surfaces of the components of the energy storage system; and to interact with the lithium battery cells themselves. Thus, the system **100** can minimize an amount of fire suppression fluid required to suppress a fire and can provide fire suppression to battery storage containers **110** at locations where water and/or other fluid suppression agents are scarce, logistically difficult to coordinate, and/or prohibitively expensive to manage.
(26) The system **100** can be installed in conjunction with additional systems **100** to create a network of systems **100** that can communicate with each other to prevent fire and heat propagation between adjacent battery storage containers no and can supplement fire suppression fluid (e.g., via connecting pipes) to adjacent systems as needed to suppress a fire event at a particular battery storage container **110**. For example, a first system **100** of a first battery storage container **110** with an ongoing fire event can transmit a fire event warning prompt to a second system of an adjacent battery storage container **110** to activate a fire suppression response and facilitate cooling within the adjacent battery storage container **110** to decrease the ambient temperature and minimize thermal runaway between battery storage containers no.
(27) In one variation, the system **100** can: actively detect a potential fire event by monitoring outputs of multiple sensors—such as a light sensor, a humidity sensor, gas sensor, and/or a temperature sensor- and detecting changes in such measured ambient conditions; and, in response to detecting these changes, initiate a fire suppression response to cool the ambient environment within the battery storage container no and prevent the potential fire event.
(28) In another implementation, the system **100** can detect a precursor condition **102** to an incipient explosion event, such as unexpected presence of a volatile gas during nominal operation of the system **100**, or an increased concentration of a volatile gas known to be present during nominal operation at a lower concentration. In response to detecting the potential for an explosive event in the battery storage container no, the system can trigger a door **115** or vent to open, venting the gas in the battery storage container no to the external atmosphere, and reducing the potential for an explosion. The system can include explosion mitigation systems independent of, or in conjunction with, fire detection and mitigation systems.
(29) Generally, the system **100** is configured to detect a precursor condition **102** indicative of an incipient fire or explosion event particular to lithium-ion battery cells. The system **100** detects the precursor condition **102** at a time prior to development of a fire or explosion to execute a response to mitigate the precursor condition **102** and/or interrupt progression of the precursor condition **102** to a fire or explosion event. However, the system **100** can be configured to detect a precursor condition **102** indicative of an incipient fire event, explosion event, adverse chemical interaction, leak, and/or other event that may preempt a fire or explosive event and initiate an action to mitigate the precursor condition **102** and/or interrupt progression of the precursor condition **102** to a more destructive event.
(30) 3. Housing
(31) In one implementation as shown in FIG. **1**, the system **100** can include a housing formed of a unitary structure that defines the main body of the system **100**. The housing can define: a base; a perimeter wall extending upwardly from the base; a cover disposed over the perimeter wall; a storage chamber bounded by the base, the perimeter wall, and the cover; a first opening for connecting the cooling channel **151** positioned within the storage chamber to the nozzle **114**; and a second opening for connecting the cooling channel **151** to the fluid storage tank. Moreover, the housing can include a connector to a set of nozzles **114**, a connector to the cooling plate subsystem, a connector to a fluid (e.g., water) storage tank, a connector to a supplemental fluid storage supply, as well as electronic and/or electromechanical connections to a controller **130**, a power supply, and/or a reserve power supply. In one example, the housing can include a set of handles for lifting,

moving, and/or positioning the system **100** (e.g., during installation of the system **100**).
(32) The system **100** can further include a set of securing mechanisms arranged along a top portion of the housing, each securing mechanism configured to transiently or permanently secure the housing to an interior or exterior surface of the battery storage container **110**. In one example, the system **100** includes four securing mechanisms arranged at each corner of the base of the housing.
(33) In one variation, the system **100** can include a set of additional securing mechanisms arranged at the centers of the edges of the base of the housing. Accordingly, the system **100** can be secured to a ceiling of the battery storage container **110** such that the system **100** is configured to direct the fluid spray pattern downward onto the set of battery cells **122** enclosed within the battery storage container **110**, thereby controlling, suppressing, or extinguishing any fire below and preventing the fire from spreading.
(34) In another variation, the system **100** can include the set of securing mechanisms arranged along a side or bottom portion of the housing such that the system **100** can be secured to a perimeter wall or a bottom surface of the battery storage container **110**.
(35) 4. Cooling channel
(36) As shown in FIG. **1**, the system **100** can further include a cooling channel **151** positioned—or embedded—within the storage chamber of the housing. The cooling channel **151** can define: a proximal end connected to a port on the fluid storage tank; a distal end connected to the nozzle **114**; and a lumen disposed between the proximal end and the distal end.
(37) Alternatively, the cooling channel **151** can be embedded within a set of plates, each including: an inlet bringing fluid into the cooling plate from another cooling plate or the fluid supply; an outlet connecting the plate to another plate or returning the fluid to the fluid supply; an enclosed channel or plenum through which the fluid can flow from the inlet to the outlet, containing apertures **152** sealed with a meltable plug designed to fail at a specific temperature resulting in fluid distribution to the adjacent heated surface. Moreover, the cooling plates can have static or circulating fluid and can be monitored for pressure or flow to determine system activation.
(38) In another variation, the cooling plates (and cooling channels **151**) can connect independently to the fluid supply in series (e.g., multiple cooling plates connected via a single line to the fluid supply), and/or in parallel (e.g., multiple cooling plates independently connected via multiple lines to the fluid supply).
(39) In another variation, the system **100** can include a set of cooling channels **151** that branch from the proximal end connected to the port of the fluid storage tank, each cooling channel **151** having a corresponding distal end connected to a corresponding nozzle **114**. In this variation, a single system **100** can deliver the fluid at multiple directions within the battery storage container **110**. Alternatively, the set of cooling channels **151** can individually connect to a corresponding port on the fluid storage tank (rather than branching from the same port).
(40) In one implementation, as shown in FIG. **6**, the system **100** can include a set of cooling channels **151** arranged within an intercooler **150**, the intercooler **150** arranged within a battery tray **120**. The intercooler **150** is configured to receive fluid from a fluid supply, such as a supply manifold **157** or fluid storage tank, and circulate the fluid through the battery tray **120** to cool the set of battery cells **122** during nominal operation of the system **100**. In one variation in which a set of battery trays **120** is arranged in a set of battery racks **11**, each battery tray **120** including an intercooler **150**, the system **100** can: be configured to detect a precursor condition **102** to an incipient fire event, such as an increase in temperature, in a first battery tray **120** in the set of battery trays **120**; and, in response, increase the flow rate and/or pressure of fluid in the intercoolers **150** of the remaining battery trays **120** (not exhibiting the precursor condition **102**) to increase cooling. By increasing cooling of the battery cells **122**, the system can reduce or prevent the propagation of heat and/or developing fire conditions from the first battery tray **120** to an adjacent battery tray **120**.
(41) 5. Nozzle

(42) The system **100** can include a nozzle **114** configured to direct a fluid spray pattern within the battery storage container **110** enclosing the set of battery cells **122**. In one implementation, the nozzle **114** includes: a nozzle body; a nozzle lumen spanning between a proximal end and a distal end of the nozzle body; an inlet at the proximal end of the nozzle body and configured to fluidly connect the nozzle lumen to the distal end of the cooling channel **151**; an outlet at the distal end of the nozzle body and positioned to direct fluid in the fluid spray pattern at a set of battery cells **122** within the battery storage container **110**.
(43) In one variation, a portion of the nozzle body is inset into the first opening of the housing such that the outlet of the nozzle **114** protrudes from the housing. In one example, the system **100** can be secured to a ceiling within the battery storage container **110**, and the first opening of the housing can be located on a base of the housing such that the outlet of the nozzle **114** points downward out of the housing and perpendicular to a surface of a battery cell **122**. In a variation of this example, the first opening can be located in the perimeter wall of the housing such that the outlet of the nozzle **114** points parallel to a surface of a battery cell **122**.
(44) The nozzle **114** can be designed to produce a particular spray pattern, spray angle, volumetric flow rate, and drop size distribution of the fluid exiting the outlet of the nozzle **114**. In one implementation, the nozzle **114** can produce a volume median drop size (dv_50) of between 25 and 400 microns. In particular, the nozzle **114** can aerosolize the fluid such that the fluid behaves like a gas (i.e., suspended in air) and moves like a gas throughout the battery storage container **110** in order to deposit the fluid on vertical, horizontal, and angled surfaces, to get drawn in between the surfaces of the battery cells **122**, and to interact with the battery cells **122** themselves. The nozzle **114** can be configured to deliver the fluid within the enclosure by generating droplets of an ideal drop size distribution (dv_50) and surface to volume ratio (d_32) via mechanical, pneumatic, or alternative drop formation techniques.
(45) The nozzle **114** body can be configured to include multiple, different geometries, such as: a spiral nozzle, a convergent cone nozzle, and/or a flat fan nozzle. For example, the nozzle **114** can include a fire protection nozzle deflector design or multiple ejection ports to control spray dispersion. The nozzle body can also be configured to include similar variations in order to produce a hollow cone spray pattern, a jet spray pattern, a plume spray pattern, similar variations, or some combination thereof. The nozzle lumen can be configured to include multiple, different geometries such as: a varying cross-sectional area, a uniform cross-sectional area across the length of the nozzle body; and/or a set of vanes configured to cause turbulence within the nozzle lumen and atomize the fluid passing through the nozzle lumen.
(46) In one variation, the system **100** can include a set of nozzles **114**—of the same design or of varying design—configured to increase a volumetric flow rate of fluid into the battery storage container no, thereby enabling an accelerated fire suppression response within the battery storage container no.
(47) 6. Sensing & Activation
(48) The system **100** can be configured to passively and/or actively respond to changes of the ambient environment (e.g., temperature, gas, humidity, light) within the battery storage container no and either passively or actively initiate a fire suppression response. For example, the system can be activated by detecting gas production, general smoke production, and/or specific gas constituents. Moreover, the system can also include a user interface (e.g., emergency switch or trigger) configured to activate in response to user input.
(49) 6.1 Precursor Conditions
(50) As shown in FIGS. **1-2** and **4-7**, the system **100** can be configured to identify precursor conditions **102** indicating a potential or incipient fire or explosive event. In particular, the system **100** can be configured to detect a precursor condition **102** indicating an incipient fire event unique to a particular battery cell type, such as a lithium-ion battery cell, based on the characteristics of the battery cell type (such as a particular temperature threshold for a particular cell type.) Generally the

system **100** can detect a precursor condition **102** in a particular battery tray **120** and/or be configured to detect the precursor condition **102** in a particular battery cell. The controller **130** can be configured to initiate a particular mitigation action in response to detection of the precursor condition **102**, to suppress the precursor condition **102** in the battery tray **120**, and thereby mitigate or prevent a fire from developing in the battery tray **120** and/or propagating to an adjacent battery tray **120**.

(51) In one variation, the system **100** can be configured to detect a precursor condition **102** to a fire event based on a detected temperature of the battery tray **120** compared to a threshold temperature. The controller **130** can be programmed with a threshold temperature greater than or equal to a nominal operating temperature of the battery cell **122** and less than or equal to an ignition temperature of the battery cell. A sensor **113** arranged within the battery tray **120** can be configured to detect the temperature of the set of battery cells **122** in the battery tray **120** in real time and, in response to the temperature of the battery tray **120** exceeding the threshold temperature, the sensor **113** can transmit a signal to the controller **130**. In response to receiving the signal, the controller **130** can initiate a response action to suppress the elevated temperature in the battery tray **120**, such as increasing cooling or releasing fluid into the battery tray **120**. In one variation, a sensor **113** is arranged in contact with or proximal a particular battery cell **122** and configured to detect the temperature of that battery cell.

(52) In one example in which the sensor **113** is configured to detect a temperature of a battery tray **120** and transmit a signal to a controller **130**, the controller **130** can be configured to receive the signal from the sensor **113** and detect the temperature of the battery tray **120** exceeding a threshold temperature based on the signal, indicating a precursor condition **102** for an incipient fire event has been met for the set of battery cells **122** occupying the battery tray **120**. In response, the controller **130** triggers a tray ejector **124** to transition the battery tray **120** from an inserted position to an extended position; and triggers a nozzle **114** arranged in the battery tray **120** to direct fluid into the battery tray **120** to suppress the precursor condition **102**.

(53) The system **100** can also be configured to detect a precursor condition **102** to a fire event in a battery cell **122** based on a detected pressure of the battery cell. An increase in pressure in the battery cell case of a battery cell **112**, such as that caused by swelling in the battery cell case due to outgassing in the battery cell **112**, can indicate failure of the battery cell **112** without an increase in temperature. For example, the sensor **113** can be configured to detect a pressure of the battery cell case of a particular battery cell **112**, in the set of battery cells, occupying the battery tray **120**. The controller **130** is then configured to: receive a signal from the sensor **113** and detect the pressure of the particular battery cell **122** exceeding a threshold pressure based on the first signal, indicating a precursor condition **102** for an incipient fire event has been met in the battery tray **120**. In response to detection of the precursor condition **102** in the battery tray **120**, the controller **130** triggers the tray ejector **124** to transition the first battery tray **120** from the inserted position to the extended position; and triggers the first nozzle **114** to direct the fluid into the first battery tray **120** to suppress the precursor condition **102**.

(54) The controller **130** can be programmed with a threshold pressure greater than or equal to a nominal operating pressure of the battery cell case. The sensor **113** can be arranged within the battery tray **120** or proximal a particular battery cell **122** and can be configured to detect the pressure of the battery cell case. In response to the pressure of the battery cell case of battery cell **122** exceeding the threshold pressure, the sensor **113** transmits a signal to the controller **130**. In one variation, the sensor **113** can define: a pressure sensor arranged proximal the battery cell, (e.g., attached to the outer surface of the cell), a piezoelectric sensor interposed between a battery cell **122** and a rigid surface (e.g., a second, adjacent battery cell, a sidewall of the battery tray **120**), a belt arranged around the battery cell **122** and configured to detect an increase in belt tension caused by a bulge in the battery cell case, or any other sensor configured to detect material stress, fatigue, and/or deformation in the battery cell case. In response to receiving a signal indicating a pressure

increase, the controller **130** initiates a response action to isolate the battery tray **120** from adjacent battery trays **120**, such as ejecting the battery tray **120**.

(55) Therefore, the system **100** can be configured to detect various precursor conditions **102** preceding a fire event based on the battery cell type. In particular, the system **100** can be configured to detect a temperature increase beyond a temperature threshold and/or a pressure increase beyond a pressure threshold indicating various failure modes of a battery cell, such as thermal runaway or battery cell case deformation or rupture. The system can implement fire mitigation actions to prevent heat propagation through the battery storage container **110** or to arrest or slow thermal runaway of the battery cell.

(56) 6.2 Passive Sensing and Activation

(57) In one implementation, the system **100** can include a passive sensing and activation system for: detecting a probable fire within the battery storage container **110**; and activating the system **100** in response to detecting the probable fire in order to suppress the fire and/or mitigate the spread of the fire. In particular, the system **100** can include a meltable plug arranged over each of the apertures **152** disposed within the cooling channel **151**. The meltable plug **153** can be configured to melt and expose the aperture **152** in response to an ambient temperature surrounding the system **100** exceeding a threshold temperature. More specifically, the meltable plug **153** can be composed of a thermoplastic material that has a melting temperature at or near a minimum temperature of an active fire (e.g., approximately 200 degrees Celsius). Accordingly, in response to the ambient temperature approaching or exceeding 200 degrees Celsius, the meltable plug can melt and expose the aperture, thereby enabling the passage of fire suppression fluid from the fluid storage tank, through the cooling channel **151**, thus directing the flow of fire suppression fluid toward the set of battery cells **122** within the battery storage container **110**. Accordingly, the system **100** can discharge the fire suppression fluid in a fluid spray pattern toward the set of battery cells **122** in response to a specific change in ambient temperature indicative of an active fire within the battery storage container **110**.

(58) For example, as shown in FIG. **6**, the cooling channel **151** can: define a set of apertures **152** configured to release fluid from the cooling channel **151** into the first battery tray **120**; and include a set of meltable plugs **153**, each meltable plug in the set of meltable plugs **153** configured to insert into an aperture in the cooling channel **151**, seal the aperture when a temperature of the first battery tray **120** is maintained below a threshold temperature, thereby retaining fluid within the cooling channel **151**, and melt in response to the temperature in the first battery tray **120** exceeding a threshold temperature, thereby releasing the fluid into the first battery tray **120**.

(59) Therefore, the system **100** can cool the set of battery cells **122** within a battery tray **120** under nominal operating conditions, and direct fluid into the battery tray **120** in response to a precursor condition **102** utilizing the intercooler **150**.

(60) In a variation of this example, the meltable plugs **153** are resistant to high pressure and will expose the aperture **152** in the cooling channel **151** when exposed to an elevated temperature, but not when exposed to an elevated pressure in the cooling channel **151**. In this variation, the intercoolers **150** including meltable plugs **153** are arranged in each battery tray **120** in a set of battery trays **120**. A first battery tray **120** exhibits a precursor condition **102** defining an elevated temperature and, in response, the meltable plugs **153** in the apertures **152** of the intercooler **150** in the first battery tray **120** melt, releasing fluid into the first battery tray **120**. The system **100** increases the fluid pressure in the remaining intercoolers **150** to increase cooling within the remaining battery trays **120** not exhibiting elevated temperatures, without ejecting the meltable plugs **153**. The increased cooling in the remaining battery trays **120** slows or prevents propagation of heat from the first battery tray **120** exhibiting the elevated temperature to an adjacent battery tray **120**. In the event the heat propagation overwhelms the cooling in the adjacent battery tray **120**, the meltable plugs **153** melt, exposing the apertures **152** of the cooling channel **151**. Fluid is then directed into the adjacent battery tray **120** via the exposed apertures **152** to further increase cooling

in the adjacent battery tray **120**.

(61) In another implementation, the system **100** can further include a passive electrical disconnect **140** configured to sever an electrical connection between a power bus **142** and the set of battery cells **122** in response to ejection of the battery tray **120**. Electrically disconnecting a battery cell **122** exhibiting a precursor condition **102** mitigates thermal runaway in the battery cell **122** by preventing additional current flowing to the battery cell, thereby reducing the rate of temperature increase in the battery cell. Electrically disconnecting the battery cell **122** further reduces or eliminates the possibility of short circuits and/or other injury to additional electrical circuits or other battery cells **122** due to interaction between live electrical circuits and fire suppression fluid, particularly when the battery cell **122**, exhibiting the precursor condition **102**, is immersed in or exposed to fire suppression fluid, such as when fire suppression fluid is directed into the battery tray **120**.

(62) In one example as shown in FIG. **6**, the system **100** can include: a power bus **142** arranged proximal the battery rack **111**; and an electrical disconnect **140** electrically coupled to and interposed between the power bus, and the set of battery cells **122** within a battery tray **120**. The electrical disconnect **140** can be configured to: electrically couple the power bus **142** to the set of battery cells **122** in a coupled state; physically and electrically disconnect the power bus **142** from the set of battery cells **122** in a decoupled state, thereby isolating the battery cell; and transition from the coupled state to the decoupled state in response to ejection of the battery tray **120** from the battery rack **111**.

(63) In one variation of this example, the electrical disconnect **140** can define a set of electrical contacts configured to repeatably and non-destructively couple and decouple. The set of electrical contacts can include: a first electrical contact arranged proximal and electrically coupled to the power bus **142** and configured to remain stationary when the electrical disconnect **140** is in both a coupled and decoupled state; and a second electrical contact affixed to the battery tray **120**, electrically coupled to the set of battery cells **122** occupying the battery tray **120**, and configured to insert into the first electrical contact with an interference fit in the coupled state, thereby completing an electrical connection between the power bus **142** and the set of battery cells **122**. When transitioning from the coupled state to the decoupled state, the second electrical contact is configured to release from the first electrical contact as the battery tray **120** is ejected from the battery rack **111**, severing the electrical connection. In one variation, the electrical disconnect **140** is reusable. After a fire event occurs, causing the battery tray **120** to be ejected from the battery rack **111**, the failed set of battery cells **122** is removed and replaced with a new set of battery cells **122** and electrically coupled to the second electrical contact. As the battery tray **120** is re-inserted into the battery rack **111**, the second electrical contact inserts into the first electrical contact, establishing an electrical connection between the power bus **142** and the new set of battery cells **122**. In another variation, the set of electrical contacts can couple via methods other than interference fit (e.g., friction fit, magnetic coupling, and/or push-pin contact).

(64) In yet another implementation, as shown in FIG. **4**, the system **100** can include a passive tray ejector **124** in combination with the intercooler **150** including meltable plugs **153** and the passive electrical disconnect **140** to define a passive system configured to detect a precursor condition **102** and automatically execute a set of mitigation steps to suppress the detected precursor condition **102**. The passive tray ejector **124** can include: a spring fixed to the battery rack **111** at a distal end and affixed to the battery tray **120** at a proximal end, configured to compress (i.e., load) when the battery tray **120** is in the inserted position in the battery rack **111**, and extend (i.e., unload) when the battery tray **120** is in the extended position external to and supported by the battery rack **111**. The passive tray ejector **124** can further include a thermally-sensitive tray latch (e.g., a latch configured to fail in response to an increased temperature, a passive thermocouple electrically coupled to an electromechanical release) configured to retain the battery tray **120** in the inserted position when the ambient temperature proximal the tray latch is below a threshold temperature and release the

battery tray **120** in response to the ambient temperature proximal the tray latch exceeding the threshold temperature. In response to the ambient temperature exceeding the threshold temperature, the tray latch releases the battery tray **120** and the spring extends, ejecting the battery tray **120** from the battery rack **111**. In another variation, the passive ejector can include the thermally-sensitive tray latch and the battery tray **120** arranged at a downward angle. When the tray latch disengages, the battery tray **120** slides out of the battery rack **111** under the force of gravity. When the passive tray ejector **124** is combined with the intercooler **150** including meltable plugs **153** and the passive electrical disconnect, the system **100** defines a system configured to passively detect an increase in temperature of a battery cell **122** above a threshold temperature, and automatically execute a set of mitigation steps to suppress the elevated temperature in the battery tray **120** including cooling the battery cell, directly exposing the battery cell **122** to fire suppression fluid, severing the electrical connection to the battery cell, and isolating the battery tray **120** from adjacent battery trays **120**.

(65) Therefore, the system **100** can passively: detect a precursor condition **102** in a set of battery cells **122** occupying a battery tray **120**; and, in response to detection of the precursor condition **102**, automatically execute mitigation steps to suppress the precursor condition **102**. The system **100** can combine passive systems to: detect a precursor condition **102**, cool a battery cell, expose a battery cell **122** directly to fire suppression fluid, sever an electrical connection to the battery cell, and isolate a battery tray **120**. Passive systems can be combined to define an automatic detection and mitigation system requiring no power or input; and/or be combined with active systems to provide redundancy to active systems and/or reduce complexity by replacing active systems requiring power or input.

(66) 6.3 Active Sensing and Activation

(67) The system **100** can include an active sensing and activation system for: detecting a probable fire within the battery storage container **110**; and activating the system **100** in response to detecting the probable fire in order to suppress the fire and/or mitigate the spread of the fire. In particular, the system **100** can include: a sensor **113** (e.g., a temperature, humidity, and/or light sensor **113**) configured to detect a change in ambient conditions within the battery storage container **110**; and an electro-mechanical valve (e.g., a directional control valve **156**, a bi-stable valve) configured to open and release fire suppression fluid to flow from the fluid storage tank, through the cooling channel **151**, and through the nozzle **114** directing the flow of fire suppression fluid toward the set of battery cells **122** within the battery storage container **110**. In this implementation, the controller **130** can be configured to activate the electro-mechanical valve to open in response to the sensor **113** detecting the change in ambient conditions within the battery storage container **110** exceeding a threshold (e.g., a change in temperature exceeding 100 degrees Celsius). The electro-mechanical valve can be positioned in-line at the proximal end of the cooling channel **151** near the fluid storage tank or at the distal end of the cooling channel **151** near the nozzle **114**. When inactive, the electro-mechanical valve can remain in a default closed state to prevent the passage of fluid through the system **100**. Accordingly, the system **100** can discharge the fire suppression fluid in a fluid spray pattern directing the flow of fire suppression fluid toward the set of battery cells **122** in response to detecting a fire within the battery storage container **110**.

(68) 6.3.1 Overhead Sensor

(69) In one implementation, the system **100** can include a sensor **113** arranged within the battery storage container **110** external to the battery racks **111** and battery trays **120**. The sensor **113** can define an infrared sensor, thermal imaging sensor, remote temperature sensor, light sensor, or another sensor. The sensor **113** can be configured to detect global ambient conditions within the battery storage container **110**, or be configured to detect conditions in a particular battery rack **111** or battery tray **120** within the battery storage container **110**.

(70) In one example, the system **100** can include the sensor **113** arranged external to a battery rack **111**, such as arranged on a ceiling of the battery storage container **110** and configured to detect a temperature within a battery tray **120** inserted into the battery rack **111**. The battery tray **120**

includes a tray ejector **124** configured to transition the battery tray **120** from an inserted position to an extended position. The controller **130** is further configured to: receive a signal from the sensor **113**; detect the precursor condition **102** in the battery tray **120** based on the signal; and, in response to detecting the precursor condition **102** in the battery tray **120**, trigger the tray ejector **124** to transition the battery tray **120** from the inserted position to the extended position. In one variation of this implementation, the sensor **113** is an infrared sensor capable of capturing infrared images.
(71) In one variation, a set of overhead sensors **113** are placed within the battery storage container **110** to provide an overlapping field of view, and/or redundancy to sense precursor conditions **102** within the battery storage container **110**. In another variation, a set of overhead sensors can be installed in addition to sensors **113** arranged within battery trays **120** to supplement detection of precursor conditions **102**.
(72) Therefore, a single sensor **113** or array of sensors can be arranged within the battery storage container **110** external to the battery trays **120** and battery racks **111**, such as in an overhead configuration, to detect precursor conditions **102**. The external sensor can be a simple temperature sensor configured to detect ambient conditions within the battery storage container **110**, or a thermal imaging sensor configured to detect a precursor condition **102** in an individual battery tray **120**. The external sensor can be installed as a retrofit in an existing battery storage container **110**, or as a supplement to individual sensors arranged within battery trays **120**.
(73) 6.3.2 Directional Control Valve
(74) In one implementation as shown in FIG. **1**, the system **100** can include a directional control valve **156** fluidly coupled to and interposed between: a supply manifold **157** configured to supply fluid and an intercooler **150**; and the supply manifold **157** and a nozzle **114**. The directional control valve **156** is configured to: in a first state, receive fluid from the supply manifold **157** and direct fluid to the intercooler **150**; and, in a second state, receive fluid from the supply manifold **157** and direct fluid to the nozzle **114**. In this implementation, triggering the nozzle **114** to direct fluid into the battery tray **120** is effected by controlling the directional control valve **156** to transition from the first state to the second state, thereby directing fluid from the supply manifold **157** to the nozzle **114**.
(75) The directional control valve **156** enables the system **100** to actively change the routing of fluid from the intercooler **150** to the nozzle **114** in response to a detected precursor condition **102**. In one implementation, wherein a set of battery cells **122** requires active cooling, the directional control valve **156** can direct fluid to the intercooler **150** to maintain the set of battery cells **122** at a nominal operating temperature at a first time. At a second time, in response to an elevated temperature detected in the tray, the controller **130** can control the directional control valve **156** to transition to the second state directing the fluid into the battery tray **120** via the nozzle **114** to cool the set of battery cells **122**. The directional control valve **156** can be controlled independently of or in conjunction with ejection of the battery tray **120**. In one variation, the directional control valve **156** can be controlled to switch from directing fluid to the intercooler **150** to directing fluid to the nozzle **114** in response to ejection of the battery tray **120**, as opposed to detection of the precursor condition **102**.
(76) In one variation in which the first battery tray **120** is configured to extend from the battery rack **111** and receive fluid, the first battery tray **120** further includes a seal, arranged at an end of the first battery tray **120** proximal the battery rack **111** when the first battery tray **120** is in the extended position, and configured to prevent fluid from passing through a gap between the first battery tray **120** in the extended position and the battery rack **111**, thereby preventing fluid from escaping the first battery tray **120** and flowing into a second battery tray **120** arranged below the first battery tray **120** in the battery rack **111**. The seal prevents fluid passing through the gap to reduce or prevent potential damage to other sets of battery cells **122** in other battery trays **120** arranged within the battery rack **111** caused by interaction with the fluid, and enables the system **100** to maintain electrical connection to other sets of battery cells **122** in other battery trays **120** in

the battery rack **111** not exhibiting the precursor condition **102**, thereby reducing the risk of short circuit due to electrical interaction between electrical components and the fluid, and preserving a greater amount of energy storage capacity of the system **100**.

(77) 6.2.4 Secondary Pump and Active Fluid Release

(78) In one implementation as shown in FIG. **6**, in which the system **100** includes an intercooler **150** configured to circulate fluid through the battery tray **120**, the intercooler **150** further includes: a cooling channel **151** defining a set of perforations configured to release fluid from the cooling channel **151** into the first battery tray **120**; and a set of pressure sensitive plugs **154**. Each pressure sensitive plug **154** is configured to: insert into a perforation in the cooling channel **151**; seal the perforation when a pressure of fluid in the cooling channel **151** is below a threshold pressure, thereby retaining the fluid in the cooling channel **151**; and eject from the perforation in response to the pressure of fluid in the cooling channel **151** increasing to greater than the threshold pressure, thereby releasing fluid into the first battery tray **120**. In one variation of this implementation, the system can further include a dedicated pump, fluidly coupled to the intercooler **150**, and configured to increase the pressure of fluid in the cooling channel **151** above the threshold pressure in response to detection of the precursor condition **102**. The controller **130** can be configured to trigger the dedicated pump **155** to activate, thereby increasing the pressure in the cooling channel **151** above the threshold pressure and ejecting the set of pressure sensitive plugs **154** in response to a signal received from a sensor **113**.

(79) Therefore, the system **100** can cool the set of battery cells **122** within a battery tray **120** under nominal operating conditions, and direct fluid into the battery tray **120** in response to a precursor condition **102** utilizing the intercooler **150** without a secondary component.

(80) 6.3.5 Active Electrical Disconnect

(81) In one implementation, as shown in FIG. **6**, the system **100** includes an active electrical disconnect **140** configured to disconnect a set of battery cells **122** exhibiting a precursor condition **102** and/or a fire event from additional electrical circuits in the system **100**. The active electrical disconnect **140** can define a relay, transistor, or other electrically operable switch capable of severing an electrical connection in response to receiving a signal from the controller **130**. Generally, the active electrical disconnect **140** is a relay configured to electrically isolate the power bus **142** from the set of battery cells **122** exhibiting the precursor condition **102**. However, in an implementation in which a faster switching speed is required, the electrical disconnect **140** can be a transistor or other suitable electrically operable switch. Actively disconnecting the battery cell **122** exhibiting a precursor condition **102** can be implemented independent of other mitigation actions (e.g., ejecting the battery tray **120**, directing fluid into the battery tray **120**) to mitigate thermal runaway in the battery cell **122** by preventing additional current flowing to the battery cell, thereby reducing the rate of temperature increase in the battery cell.

(82) In one example, the system **100** includes the power bus **142** arranged in the battery rack **111** adjacent the battery tray **120**, and an electrical disconnect **140** interposed between the power bus **142** and the set of battery cells **122** within the battery tray **120** configured to: electrically couple the power bus **142** to the set of battery cells **122** in a coupled state; electrically disconnect the power bus **142** from the set of battery cells **122** in a decoupled state; and transition from the coupled state to the decoupled state. The controller **130** is configured to trigger the electrical disconnect **140** to transition from the coupled state to the decoupled state in response to detection of the precursor condition **102** in the first battery tray **120**. In this example, the electrical disconnect **140** remains physically coupled to the power bus **142** and the set of battery cells **122** in the battery tray **120**, but severs the electrical connection, electrically isolating the power bus **142** from the set of battery cells **122**. In one variation in which the set of battery cells **122** in the tray is replaced with a new set of battery cells **122**, the electrical disconnect **140** can be reset (i.e., recoupled) when the set of battery cells **122** is replaced, as opposed to replacing a destructive electrical disconnect **140** in addition to replacing the set of battery cells **122**. Therefore the electrical disconnect **140** can be re-

used, reducing the number of components necessary to replace following execution of a mitigation action. In another variation, the electrical disconnect **140** can be controlled by the controller **130** to electrically re-couple the power bus **142** to the set of battery cells **122** in the event the controller **130** detects a false-positive precursor condition **102**. Therefore the controller **130** can automatically restore electrical storage capacity of the system in the event a false-positive precursor condition **102** is detected, and the false-positive characteristic is verified by the controller **130**, such as by comparing the false-positive signal to signals from other sensors **113** indicating nominal conditions.
(83) In another example, the controller **130** can be configured to, at a first time, detect a precursor condition **102** in a battery tray **120** based on a first signal received from a sensor **113** arranged in the battery tray **120**, the first signal indicating an increase in temperature within the battery tray **120** exceeding a first threshold temperature. In response to detecting the precursor condition **102**, the controller **130** can trigger the electrical disconnect **140** to sever the electrical connection between the power bus **142** and the set of battery cells **122**. The controller **130** can then, at a second time following the first time, receive a second signal from the sensor **113** indicating the temperature within the battery tray **120** falling below a second threshold temperature and based on the second signal, detect that the precursor condition **102** is suppressed within the first battery tray **120**. Alternatively, at the second time, the controller **130** can receive a third signal from the sensor **113** indicating the temperature in the battery tray **120** exceeding a third threshold temperature greater than the first threshold temperature and, in response, detect that thermal runaway is occurring in the battery cell **122** and/or a fire is imminent in the battery tray **120**. In response, the controller **130** can initiate additional mitigation actions such as: triggering the battery tray **120** to eject from the battery rack **111**; controlling the directional control valve **156** to switch states to direct fluid to the nozzle **114**; and/or triggering the overhead fire suppression system to direct fluid into the battery tray **120**.
(84) 7. Fluid Storage Tank
(85) In one implementation, the system **100** can include a fluid storage tank that stores a volume of fluid configured to suppress a fire and/or mitigate a spread of the fire when the fluid is sprayed within the battery storage container **110**. As used herein, the term fluid can include liquids or gases that can be stored and dispersed in liquid or gaseous/vapor form. The fluid can include water, a clean agent (e.g., an inert gas, a combination of inert gases), aerosol, a water-based solution including a water additive, or some combination thereof that can be dispersed in a vapor or gaseous form throughout the battery storage container **110**.
(86) Generally, the system **100** can include a valve for connecting directly to a water supply source provided by a fire department. In this variation, the fire department can provide an additional source of water (e.g., via a fire truck dispatched to a location of the battery storage container **110** in response to a fire alert and/or a nearby fire hydrant), and the system **100** can direct the water within the battery storage container **110**, thus providing direct access to the source of the fire. Moreover, the system **100** can connect directly to local or private water mains or complementary sources associated with the building in which the battery storage container no is located.
(87) In a variation, the system **100** can draw fluid from a set of fluid storage tanks between adjacent containers by, for example, connecting a set of tubes from the fluid storage tanks of a second system **100** on a second battery storage container no and a third system **100** on a third battery storage container no to the fluid storage tank of the first system **100**. Accordingly, the system **100** can draw additional fluid as needed from fluid storage tanks attached to containers without a current fire and redirect the fluid toward the battery storage container **110** with a current fire.
(88) 8. Power & Pump
(89) The system **100** can further include a pump **155** connected to the fluid storage tank and configured to draw fluid out of the fluid storage tank, into the cooling channel **151**, and out of the outlet of the nozzle **114**. More specifically, system **100** can include a fluid pump **155** (e.g., a water pump) and a separate fan to create a partial vacuum and draw fluid out of the fluid storage tank.

Generally, the pump **155** can supply the nozzle **114**(*s*) with fluid from the fluid storage tank and the cooling channel **151** with fluid from the fluid storage tank in response to system activation. Moreover, the pump **155** can also circulate the fluid through the cooling channel **151** to disperse heat (e.g., active fluid cooling) from adjacent cells.
(90) In one variation, the system **100** can draw fluid out of the fluid storage tank by relying on pressure differentials between the fluid storage tank and the ambient atmosphere. For example, the fluid storage tank can be pressurized, such that, when the nozzle **114** outlet is opened—via a valve release or a melted plug—the fluid can be drawn toward the lower pressure environment within the battery storage container **110** and automatically disperse fluid in vapor or gaseous form from the fluid storage tank and out of the nozzle **114** outlet.
(91) In addition, the system **100** can include a power source configured to provide power to the pump **155** and/or the controller **130**. Alternatively, the system **100** can include an electrical connection configured to connect the system **100** to the set of battery cells **122** within the battery storage container **110** in which the system **100** is installed.
(92) 8.1 Integration with Overhead Fire Suppression System
(93) In another implementation, as shown in FIG. **5**, the system **100** can be integrated into a battery storage container **110** equipped with an overhead fire suppression system. The battery rack **11** can be arranged to eject a battery tray **120** from the battery rack in and into a position beneath a fire suppression nozzle **114** arranged overhead.
(94) For example, as shown in FIG. **5**, the system **100** can include a nozzle **114** arranged above and laterally offset from the battery rack in and configured to direct fluid into the first battery tray **120** when the battery tray **120** is in the ejected position. At a first time, a sensor **113** is configured to detect a precursor to an incipient fire condition. At a second time following the first time, the first battery tray **120** is configured to eject from the first battery rack **111**. At a third time following the second time, the second nozzle **114** is configured to direct fluid into the battery tray **120**.
(95) The controller **130** executes the preceding sequence of mitigation actions to utilize the existing fire suppression system in the battery storage container **110** to suppress a precursor condition **102** in a battery tray **120**. The controller **130** can also limit damage to the remainder of the system **100** by positioning only the battery tray **120** exhibiting the precursor condition **102** beneath a nozzle **114**, thereby only exposing that battery tray **120** to a fire suppression fluid.
(96) In another implementation in which the controller **130** detects a rapid propagation of a precursor condition **102** between battery trays **120**, the controller **130** can trigger alternating battery trays **120** to eject from the battery racks **111** to maximize the distance between adjacent battery cells **122**, and trigger the existing fire suppression system to activate, thereby suppressing the propagation of the precursor condition **102** through the battery storage container **110**. Additionally, the controller **130** can cut power to the battery storage container **110**. Alternatively the controller **130** can trigger the electrical disconnect **140** of each battery tray **120** to sever the electrical connection between the power bus **142** and the set of battery cells **122** in the battery tray **120**.
(97) In one variation in which the fire suppression system utilizes an inert gas as a fire suppression fluid, the inert gas can supplement the ejection and electrical isolation of the battery trays **120** to suppress and arrest the propagation of precursor condition **102** in the battery trays **120**. By executing non-destructive mitigation actions (e.g., ejecting battery trays **120**, electrically isolating battery trays **120**, releasing an inert gas fire suppression fluid) independently or in combination to suppress the precursor condition **102** or a fire event, the system **100** can execute mitigation actions that affect all battery cells within the container, without causing collateral damage to the remaining battery trays **120** not exhibiting the precursor condition **102**. Therefore, the system **100** can execute non-destructive mitigation techniques that target and suppress a precursor condition **102** or fire event within the battery storage container **110**, while preventing collateral damage due to mitigation actions in system components not directly damaged by battery cell failure or a fire event.
(98) Therefore, the system **100** can be integrated into a battery storage container **110** equipped with

an existing fire suppression system. The controller **130** can be configured to activate individual nozzles **114** or extend individual battery trays **120** into position to receive fluid from overhead nozzles **114**. The system **100** can utilize non-destructive mitigation actions to reduce or eliminate the number of components damaged as a consequence of the mitigation action, rather than the fire event, and reduce the number of components necessary to replace following a fire suppression event to only the components directly damaged by battery cell **122** failure or the fire event itself.
(99) 9. Controller
(100) As noted above, in one variation, the system **100** can include a controller **130** configured to: monitor ambient conditions within the battery storage container **110** based on sensor **113** data; activate a fire suppression response in response to a possible fire event; release latching mechanisms; and generate and transmit a set of warning prompts associated with the fire event to a remote monitoring system and/or a fire department.
(101) 9.1 Sensor Integration and Signal Interpretation
(102) Generally as shown in FIG. **1**, the controller **130** is configured to: receive a signal from a sensor **113** arranged in the battery storage container no and interpret the signal to detect a precursor condition **102**. The controller **130** is configured to initiate actions to mitigate or suppress the precursor condition **102** in a particular battery tray **120**, set of battery trays **120**, or in the battery storage container no as a whole in response to the signal (or signals) received.
(103) In one example, the system **100** includes: a first sensor **113**, arranged in the battery rack **111** adjacent a first battery tray **120**, configured to detect a first precursor condition **102** to an incipient fire condition in the first battery tray **120**; and a second sensor **113**, arranged in the battery rack **111** adjacent a second battery tray **120**, configured to detect a second precursor condition **102** in the second battery tray **120**. The controller **130** is configured: to receive a first signal from the first sensor **113** and a second signal from the second sensor **113**; and to detect the precursor condition **102** in the first battery tray **120** based on the first signal. In response to detecting the precursor condition **102** in the first battery tray **120**, the controller **130** is configured to trigger the first tray ejector **124** to transition the first battery tray **120** from the inserted position to the extended position and trigger the first nozzle **114** to direct the fluid into the first battery tray **120** to suppress the precursor condition **102** in the first battery tray **120**.
(104) In one variation, the controller **130** can receive the second signal from the second sensor **113** and detect nominal operating conditions in the second battery tray **120** based on the second signal but, in response to detecting the precursor event in the first battery tray **120**, increase cooling in the second battery tray **120** to prevent the precursor condition **102** propagating to the second battery tray **120**.
(105) Therefore, the system **100** can define an active system including a controller **130** configured to receive and interpret a set of signals from a set of sensors **113** arranged within the battery storage container **110**, interpret the set of signals to detect presence or propagation of a precursor condition **102** in a particular location in the battery storage container **110**, and initiate a targeted response action based on the precursor condition **102**. The controller **130** can be configured to trigger additional system components to effect increasingly potent mitigation actions in the battery storage container **110**.
(106) 9.2 Monitoring and Activation
(107) In another variation, the controller **130** can be configured to activate the valve to open and activate the pump **155** in response to the sensor **113** detecting the change in ambient conditions within the battery storage container **110** exceeding a threshold (e.g., a change in temperature exceeding 100 degrees Celsius).
(108) Thus, the controller **130** can: detect an increase in ambient temperature indicative of a potential fire event; and activate the fire suppression response in response to the detected increase in ambient temperature to prevent a possible fire event within the battery storage container **110**. Moreover, the controller **130** can: detect an increase in ambient temperature indicative of an

ongoing fire event; and activate the fire suppression response in response to the detected increase in ambient temperature to suppress the ongoing fire event within the battery storage container **110** and mitigate heat propagation to adjacent battery storage battery trays **120**, units, or containers.

(109) In one variation, the controller **130** can be integrated into the system **100** thereby forming a closed-loop control circuit within the system **100**. In another variation, the controller **130** can be remotely networked into a system **100** or a set of systems **100** such that the controller **130** can: receive multiple inputs from multiple sensor **113***s* within the set of systems **100**; and control an optimized response to a potential fire by each of a set of systems **100** in response to a sensor **113** detecting a potential fire event. For example, a remotely networked controller **130** can control a response to a sensor **113** detecting a potential fire event in a first system **100** of the set of systems **100** by routing fluid resources to a container in which a potential fire event is sensed from a container(s) in which a fire event is not sensed. Additionally or alternatively, the remotely networked controller **130** can be configured as a subsystem within a larger emergency management architecture.

(110) 9.2 Isolation of Battery Tray with a Detected Precursor Condition

(111) In one variation in which battery trays **120** are arranged in the battery rack **11** in a vertical stack configuration, a battery tray **120** is configured to automatically eject from the battery rack **111** when a precursor condition **102** or fire event is detected in the battery tray **120**. The battery tray **120** is mounted on a set of slides, such as full-extension drawer slides, and is free to move horizontally beyond the edge of the battery rack **11** such that the entire volume of the battery tray **120** extends beyond the edge of the battery rack **11** while remaining physically attached. An actuator (e.g., a spring mechanism, a piston) is mounted to the set of slides and is under tension when the battery tray **120** is within the energy storage rack. The actuator extends to force the battery tray **120** to extend from the battery rack **11**. An electronically controllable, releasable latching mechanism such as an electromagnetic latch or an electrical fusible link is mounted between the battery tray **120** and the structure of the battery rack **11** and holds the battery tray **120** in place within the battery rack **11** against the expulsive force of the actuator. The controller **130** can be configured to activate the latching mechanism to release the battery tray **120** in response to a sensor **113** detecting a change in ambient conditions within the battery tray **120** exceeding a threshold (e.g., temperature, gas, humidity, light.)

(112) Upon sensing a change in the ambient conditions indicative of a precursor condition **102** or a fire event in the battery tray **120**, the controller **130** activates the latching mechanism to release, and the actuator forces the battery tray **120** to extend horizontally out of the battery rack **111** along the set of slides. Once the battery tray **120** is extended from the battery rack **111**, the battery tray **120** is isolated from the remaining battery trays **120** and the likelihood of a fire event occurring within the battery tray **120** propagating to adjacent battery trays **120** is reduced. Further, when firefighters respond to the fire event, the battery tray **120** is now exposed beyond the battery rack **111**, allowing the firefighters more freedom to respond to the fire event than if the battery tray **120** were still located within the battery rack **11**.

(113) For example, the controller **130** can detect a temperature increase at a first battery tray **120** beyond a threshold (i.e., 100 degrees Celsius) via a temperature sensor **113** mounted to the first battery tray **120**. The controller **130** can then activate the latching mechanism to release the first battery tray **120**, causing the actuator to extend and force the first battery tray **120** out of the battery rack **11**. The controller **130** can continue to monitor the temperature and other ambient characteristics of the first battery tray **120** and any other battery trays **120** proximal the first battery tray **120**. The controller **130** can transmit a warning prompt to a remote monitoring system to alert the operator of a potential and/or active fire event within the battery tray **120**, and that the battery tray **120** is extended beyond the battery rack **11**. The controller **130** can also transmit a warning prompt and real-time ambient conditions of the battery tray **120** and battery storage container **110** to firefighters. Once the firefighters arrive, the battery tray **120** is exposed beyond the battery rack

111, and the firefighters can attack the fire event from multiple angles.
(114) Thus, the controller **130** can: detect a change in the ambient conditions within or proximal the battery tray **120** indicative of a potential fire event; and activate the latching mechanism to release the battery tray **120** from the battery rack **111** in response to the detected ambient conditions in order to isolate the battery tray **120** and prevent a possible fire event within the battery rack **111**. Moreover, the controller **130** can: detect a change in the ambient conditions within or proximal the battery tray **120** indicative of an ongoing fire event; and activate the latching mechanism to release the battery tray **120** from the battery rack **111** in response to the detected ambient conditions to isolate the battery tray **120** to mitigate heat propagation to adjacent battery storage battery trays **120**, units, or containers. The controller **130** can then alert an operator or emergency responders to the potential and/or active fire event and transmit the current or last recorded conditions of the battery tray **120** and battery storage container **110**.
(115) In another variation, the system **100** can actively detect a potential fire event within a battery tray **120** by monitoring outputs of multiple sensors including a light sensor, humidity sensor, gas sensor, and a temperature sensor. The controller **130** can activate the latching mechanism to release in response to a sensor **113** detecting a change in ambient conditions within or proximal the battery tray **120** that precede a fire event, such as detecting an increase in the volume of a gas (i.e., hydrogen) beyond a threshold proximal the battery tray **120**.
(116) In another variation as shown in FIG. **4**, the battery tray **120** is mounted within the battery rack **11** on a set of slides sloping downward. When the controller **130** activates the latching mechanism to release, the battery tray **120** slides out of the battery rack in under the pull of gravity.
(117) In another variation, the battery tray **120** is mounted on a set of rails such that when the battery tray **120** is forced out of the battery rack **111**, the battery tray **120** separates from the battery rack in and falls to the ground.
(118) 9.4 Arresting Fire Propagation
(119) Generally, as shown in FIG. **6**, the system **100** is configured to implement mitigation actions to interrupt, slow, or prevent propagation of a precursor condition **102** (such as increased temperature) or a fire event from a first set of battery cells **122** to a second set of battery cells **122**. The system **100** can implement mitigation actions to suppress a precursor condition **102** detected in a first battery tray **120**, such as: severing the electrical connection to the set of battery cells **122**; and/or cooling the set of battery cells **122** directly via a nozzle **114** or indirectly via an intercooler **150** as noted above. The system **100** can execute mitigation actions in a sequence to arrest the propagation in a minimum time duration, such as first implementing a most potent mitigation action (e.g., flooding the battery tray **120** with fluid), and subsequently implementing a less potent mitigation action that causes less damage to the components of the system **100** (e.g., electrically isolating the set of battery cells **122**.) In one example, in response to receiving a first signal from a sensor **113** in a battery tray **120**, the controller **130** initiates a first mitigation action to electrically isolate the set of battery cells **122** (i.e., via the electrical disconnect) to suppress the precursor condition **102** in a particular battery cell in the set of battery cells **122**. The controller **130** is configured to execute electrical isolation as a first mitigation action to preserve the remaining battery cells **122** in the set of battery cells **122**. The controller **130** then receives a second signal from the sensor **113** and detects that the precursor condition **102** is still present in the battery tray **120** and, in response, triggers the first tray ejector **124** to eject the first battery tray **120** from the battery rack **111**. The system **100** ejects the first battery tray **120** to increase the distance between the battery trays **120** to prevent or reduce the likelihood of heat transferring between battery trays **120**, thereby interrupting propagation of heat between adjacent battery trays **120** and reducing or preventing the chance of a thermal runaway chain reaction in the battery storage container **110**. In one variation of this example wherein the first battery tray **120** continues to exhibit the precursor condition **102** after ejection, the controller **130** can additionally trigger a nozzle **114** to direct fluid into the battery tray **120** to suppress the precursor condition **102**.

(120) Therefore the system **100** can implement mitigation actions in an increasingly potent but increasingly destructive sequence to preserve the greatest number of system components while successfully suppressing a precursor condition **102** or fire event before the entire contents of the battery storage container **110** are destroyed.
(121) In one implementation, the system **100** can be configured to eject adjacent battery trays **120** in an alternating fashion to maximize the distance between adjacent battery trays **120** to mitigate the propagation of a precursor condition **102** or developing fire event. For example, the system **100** can include: a first battery tray **120** arranged in a battery rack **11** and configured to eject from the battery rack **11**; a second battery tray **120** arranged above the first battery tray **120** in the battery rack **11**; and a first sensor **113** arranged in the first battery tray **120** configured to detect the precursor condition **102** in the first battery tray **120**. In response to detection of the first precursor condition **102**, the first battery tray **120** ejects from the battery rack **11** to increase the distance between the first battery tray **120** and the second battery tray **120**, thereby reducing potential heat transfer between the first battery tray **120** and the second battery tray **120** and arresting or slowing propagation of the first precursor condition **102**. In a variation of this example in which the first battery tray **120** is fixed to the battery rack **11**, the second battery tray **120**, arranged above the first battery tray **120**, is configured to eject from the battery rack **111** in response to detection of the first precursor condition **102** in the first battery tray **120**. In another example, the preceding steps can be implemented for a first battery tray **120** and a second battery tray **120**, which are arranged laterally adjacent. Additionally, the preceding steps can be implemented for an array of battery trays **120** arranged vertically and laterally in a battery rack **111** or set of battery racks **111**, to eject alternating battery trays **120** to maximize the distance between any two adjacent battery trays **120** to arrest propagation of a precursor condition **102** or developing fire event.
(122) In one variation of this implementation, the controller **130** can selectively eject an individual battery tray **120** or a sequence of battery trays **120** to mitigate propagation of the precursor condition **102**. For example, the system **100** can include: a first battery tray **120** configured to eject from a battery rack **11** via a first tray ejector; a first nozzle **114** arranged within the first battery tray **120** configured to direct fluid into the first battery tray **120**; and a first sensor **113** configured to detect a precursor condition **102** in the first battery tray **120** at a first time. The system **100** can further include a similar second battery tray **120**, second nozzle **114**, and a second sensor **113** configured to detect a second precursor condition **102** at a second time. The system **100** can further include a similar third battery tray **120**, third nozzle **114**, and a third sensor **113** configured to detect a third precursor condition **102** at a third time.
(123) In this example, the controller **130** is configured to receive a first signal from the first sensor **113** at the first time and detect the first precursor condition **102** in the first battery tray **120** based on the first signal. In response to detecting the precursor condition **102** in the first battery tray **120**, the controller **130**: triggers the first tray ejector **124** to transition the first battery tray **120** from the inserted position to the extended position; and triggers the first nozzle **114** to direct the fluid into the first battery tray **120** to suppress the precursor condition **102** in the first battery tray **120**. The controller **130** then receives a second signal from the second sensor **113** at the second time and detects the second precursor condition **102** in the second battery tray **120** based on the second signal. In response to detecting the second precursor condition **102** in the second battery tray **120**, the controller **130**: triggers the third tray ejector **124** to transition the third battery tray **120** from the inserted position to the extended position; and triggers the second nozzle **114** to direct fluid into the second battery tray **120** to suppress the second precursor condition **102** in the second battery tray **120**.
(124) Therefore, the controller **130** can receive a first signal indicating a first precursor condition **102** in a first battery tray **120** at a first time, receive a second signal indicating a second precursor condition **102** in a second tray adjacent to the first tray at a second time, and detect propagation of a fire event within the battery storage container **110** based on the two signals and the relative

locations of the first battery tray **120** and the second battery tray **120**. Further, the controller **130** can interpret the first and second signals to predict propagation of the precursor condition **102** to the third battery tray **120**, and preemptively trigger the third battery tray **120** to eject from the battery rack **11**, thereby interrupting propagation of heat to the third battery tray **120** before it can exhibit a precursor condition **102**.

(125) 9.5 Warning Prompts

(126) In one variation, as shown in FIGS. **1** and **8**, the controller **130** can be configured to transmit a warning prompt to a remote monitoring system to alert an operator of a potential and/or active fire event within a battery storage container **110**.

(127) In this variation, the system **100** can further include a sensor **113** (e.g., a temperature, humidity, gas, and/or light sensor **113**) configured to detect a change in ambient conditions within the battery storage container **110**, wherein the controller **130** can be configured to transmit the warning prompt to the remote monitoring system in response to a sensor **113** detecting the change in ambient conditions.

(128) Additionally or alternatively, the system **100** can further include a flow detection sensor **113** configured to detect a flow of fluid through the cooling channel **151**/plate and/or the nozzle **114**, wherein the controller **130** is configured to transmit the warning prompt to the remote monitoring system in response to the flow detection sensor **113** detecting the flow of fluid through the nozzle **114**. For example, if the sensor **113** detects the flow of fluid through the nozzle **114** in a passive response configuration of the system, then the controller **130** can: identify that the plug has melted and therefore there is a significant change in temperature in the vicinity of the nozzle **114**; and, in response, transmit the warning prompt to the remote monitoring system indicating a potential fire event.

(129) Moreover, the controller **130** can be configured to: automatically alert and/or dispatch a fire department to a location of the battery storage container **110**; and/or activate a set of systems **100** on or in adjacent battery storage containers **110** (i.e., to prevent spread of an active fire event). Alternatively, the controller **130** can be configured to: transmit the warning prompt; transmit information associated with the fire event (e.g., a location of the battery storage container **110**, sensor **113** data); and display these options described above on a user interface of the remote monitoring system, thereby enabling the operator to select a set of actions based on the fire event information.

(130) For example, the system **100** can include the sensor **113** configured to: at a first time preceding ejection of the battery tray **120** from the battery rack **111**, detect a first temperature within the battery tray **120** below a threshold temperature; at a second time following the first time, detect a second temperature within the battery tray **120** exceeding the threshold temperature; and, at a third time following the second time and following ejection of the battery tray **120** from the battery rack **11**, detect a third temperature within the battery tray **120** falling below the threshold temperature. The controller **130** is configured to receive a first signal from the sensor **113** indicating the second temperature detected in the battery tray **120** at a fourth time following the second time and preceding the third time. In response, the controller **130** detects the precursor condition **102** in the battery tray **120** based on the first signal. The controller **130** receives a second signal from the sensor **113** at a fifth time following the third time, and in response, calculates a present risk value of approaching the battery tray **120** based on the second signal, the composition of the set of battery cells **122** in the battery tray **120**, and a difference between the fifth time and the present time. The controller **130** can then detect the present risk value falling below a threshold risk value. Generally the present risk value is inversely proportional to the difference between the fifth time and the present time, and as such, the present risk decreases as time progresses. In response to detecting the present risk value falling below the threshold risk value, the controller **130** is configured to: generate a prompt to approach the battery tray **120** and replace the first set of battery cells **122** in the first battery tray **120**, the prompt including the present risk value; and transmit the

notification to an operator.
(131) Therefore, the controller **130** can be configured to: receive a set of signals from a sensor **113** arranged in the battery storage container **110**; interpret the set of signals to detect and/or characterize the conditions of a battery tray **120**; calculate a risk value of a human approaching the battery tray **120** based on the conditions in the battery tray **120**, the composition of the battery tray **120** contents, and a time duration. Based on the calculated risk falling below a risk threshold, the controller **130** can generate and transmit a notification to a human operator to service the battery tray **120** when a detected risk level is below a threshold risk level.
(132) 10. System Installation
(133) The exemplary system **100** can be installed and secured within various types of battery storage containers **110**. During installation, the system **100** can be positioned within the battery storage container **110**, for example, based on an arrangement of the set of battery cells **122** within the battery storage container **110**. In one example, the set of battery cells **122** can be vertically arranged within the battery storage container no, and the system **100** can be secured to a ceiling of the battery storage container **110**, such that the system **100** directs the fluid spray pattern from the nozzle **114** downward, onto, and in between the set of battery cells **122** (e.g., to improve movement of the fluid throughout the battery storage container no and reach surfaces of the set of battery cells **122**). In another example, the set of battery cells **122** can be horizontally stacked within the battery storage container no, and the system **100** can be secured to a side wall of the battery storage container no, such that the system **100** directs the fluid spray pattern from the nozzle **114** toward a side of the set of battery cells **122** and in order to reach surfaces of the set of battery cells **122**.
(134) In one variation, the system **100** can be secured to an outside surface of a battery storage container no, for example, given limited space within the battery storage container no and/or for improved access to an external fluid storage tank. In this variation, a set of perforations can be cut into the surface of the battery storage container no during installation of the system **100**, each perforation configured to receive a nozzle **114** such that the outlet of the nozzle **114** enters an internal environment of the battery storage container **110** containing the set of battery cells **122**.
(135) The system **100** can be installed in conjunction with additional systems **100** to create a network of systems **100** that can communicate with each other in order to prevent fire and heat propagation between adjacent battery storage containers no and can supplement fire suppression fluid (e.g., via connecting pipes) to adjacent systems **100** as needed to suppress a fire event at a particular battery storage container **110**. For example, a first system **100** of a first battery storage container **110** with an ongoing fire event can transmit a fire event warning prompt to a second system **100** of an adjacent battery storage container **110** to activate a fire suppression response and facilitate cooling within the adjacent battery storage container no in order to decrease the ambient temperature and minimize thermal runaway between battery storage containers **110**.
(136) Furthermore, an individual system **100** can include a particular specification for an energy rating of a battery energy storage system (BESS)—for example, one system **100** is suitable for up to a specified number of megawatt-hours (MWh), such as 4 MWh. Accordingly, more than one system **100** can be installed on or in an individual battery storage container no in order to match the rating requirements of the set of battery cells **122** stored within the battery storage container no.
(137) 11. Battery Tray and Container Configuration
(138) In another example implementation, the system **100** can be integrated within a device and/or system containing a set of battery cells **122**. For example, the system **100** can be fully integrated within the battery storage container no enclosing the set of battery cells **122**. In this example, the battery storage container no can include: a fluid storage tank for storing a volume of fire suppression fluid and mounted within the battery storage container **110**; a cooling channel **151** mounted across an internal surface of the battery storage container no and having a lumen disposed between a proximal end and a distal end of the cooling channel **151**, the proximal end connected to the fluid storage tank; a nozzle **114** positioned at the distal end of the cooling channel **151**, the

nozzle **114** including an inlet connected to the cooling channel **151** and an outlet positioned to direct a fluid spray pattern toward the set of battery cells **122**; a meltable plug arranged over the outlet of the nozzle **114** and configured to melt and expose the outlet in response to an ambient temperature surrounding the system exceeding a threshold temperature; a flow detection sensor **113** configured to detect a flow of fluid from the fluid storage tank, into the cooling channel **151**, and out of the outlet of the nozzle **114**; and a controller **130** configured to, in response to the flow detection sensor **113** detecting the flow of fluid through the outlet of the nozzle **114**, transmit a warning prompt to a remote monitoring system. Moreover, the battery storage container no can include a set of cooling channel **151***s*—each fluidly connected to the fluid storage tank—with a corresponding set of nozzle **114***s* mounted across the internal surfaces of the battery such that the set of nozzle **114***s* can direct a fluid spray pattern at different locations (e.g., top portion, bottom portion, side surfaces, corners) within the battery storage container no. Additionally, the battery storage container no can include a valve inset into a wall of the battery storage container no such that an external fluid supply can be connected—via a pipe or hose—to supplement the fluid stored within the integrated fluid storage tank.
(139) Further, a set of battery storage containers no within an integrated system **100** can be arranged into an array of battery storage containers no within a field (i.e., a “multi-container field”). For example, a pipe or hose can connect a first valve on a first battery storage container **110** and a second valve on a second battery storage container **110**. In another example, a controller **130** of a first battery storage container no and a controller **130** of a second battery storage container no can directly or remotely communicate, such that the controller **130** of the first battery storage container no with an ongoing fire event can transmit a fire event warning prompt to the controller **130** of the second battery storage container no in order to activate a fire suppression response and facilitate cooling within the second battery storage container no. Accordingly, each battery storage container no can be linked to adjacent battery storage containers **110** to: supplement fluid to battery storage containers no with active fire events; activate the integrated fire suppression system of battery storage containers no adjacent a battery storage container no with an active fire event; and mitigate the spread of the fire and heat.
(140) 11.1 Waste Tank
(141) As shown in FIGS. **3** and **5**, The system **100** can further include an integral waste tank **160** located within the battery storage container no—such as in the bottom of the battery storage container no and between the base and the floor of the storage chamber—and configured to collect fluid runoff from fluids deposited within the storage chamber in response to a fire event.
(142) In one implementation, the system **100** includes: a set of vertical supports arranged within the bottom of the battery storage container no; and a grate arranged above the vertical supports to cover a volume in the base of the battery storage container no and to form a rigid structure configured to support other elements of the system **100** above this volume. The volume under the grate in the base of the battery storage container **110** forms the waste tank, and other elements within the battery storage container **110** can be arranged on and/or attached to the grate such that fluid runoff from a fire event within the battery storage container no drains into and is stored in this waste tank. The system can further include a pipe connection positioned on the exterior of the battery storage container **110** and connected to the waste tank, such as in the form of a sump passing through a wall of the battery storage container no above the grate and extending downwardly into the waste tank **160** to enable contents of the waste tank **160** to be pumped out of the battery storage container no and reducing opportunity of leaks (e.g., due to faulty seals or drain plug installation).
(143) In another implementation, the system **100** includes an external waste tank **160** arranged beneath, and fluidly coupled to, the battery storage container no. Generally, during installation of the system **100**, the waste tank **160** is first installed, such as bolted or otherwise affixed to a concrete pad in a field, and the container is then installed above the waste tank. In this implementation, the container can include a perforated floor **112** and the waste tank **160** can

include a perforated ceiling or an open top connected to the bottom of the container. In another implementation, the container and waste tank **160** can be fluidly coupled via a drain and a set of interconnecting pipes.

(144) By collecting the fluid runoff from the battery storage container no, the system can reduce risk of environmental contamination by fluid runoff, by-products of a fire event, or fire suppression agents (e.g., fire-retardant chemicals) consumed or deployed during a fire event in the battery storage container no. For example, the waste tank **160** can define a volume greater than a target maximum volume of fluid allocated for response to a fire event within a target quantity (e.g., three) of battery storage containers **110** or slightly (e.g., 10%) greater than a volume of fire suppression materials stored onboard the battery storage container **110**.

(145) For example, in response to detecting a fire event, the controller **130** activates the cooling system to pump **155** or deposit a volume of fluid within the battery storage container **110**. This fluid reduces temperatures of structures within the battery storage container **110**; flows downwardly —under gravity—through the grate; and collects in the waste tank **160** in the base of the battery storage container **110**. Furthermore, this fluid may collect contaminants—such as carbon, dissolved gasses, or other by-products of combustion or reaction with battery cells **122**—as this fluid flows into the waste tank. The waste tank **160** can therefore retain the contaminants away from racks, battery cells **122**, and other structures above the grate to prevent further contaminations within the battery storage container **110**. (The waste tank **160** can also be coated with a polymer or other coating robust to such contaminants.) Later, the waste tank **160** can be pumped or drained to remove this fluid.

(146) Further, the waste tank **160** (and any coating applied to the interior) can be configured to resist a temperature of a fully developed fire in the container, thereby preventing the waste tank **160** from melting, rupturing, deforming, or otherwise failing, and thereby causing the waste tank **160** to release contained fluid into the external environment. In one variation in which the contents of the container are exceptionally volatile, the waste tank **160** is configured to resist a maximum temperature of a conflagration within the container capable of consuming the entire contents of the container. In this variation, the waste tank **160** is configured to withstand a total catastrophic loss of the battery storage container **110**, without releasing solid or liquid contaminants from the waste tank **160** into the surrounding environment, thereby reducing environmental contamination when a fire event cannot be suppressed and a majority of the battery storage container **110** and its contents are destroyed.

(147) In one implementation, the waste tank **160** works in conjunction with a container including a perforated floor, and a battery tray **120** including a perforated or otherwise fluid-permeable base configured to release fluid from the battery tray **120** in order to capture contaminated fluid runoff from the battery tray **120**. Generally, after fluid is directed into the battery tray **120**, to cool the set of battery cells **122**, the perforated tray base **126** releases fluid cells into the space below the battery tray **120**, the fluid carrying heat away from the set of battery cells **122** and thereby increasing cooling. In particular, a perforated tray base **126** is applicable for a set of battery cell **122** which are best cooled by moving fluid. Fluid released through the perforated tray base **126** is then captured by the waste tank **160** arranged below the perforated floor **112** of the container.

(148) For example, the system **100** can include: a battery tray **120** including a perforated tray base **126** configured to release fluid from the battery tray **120** to carry thermal energy away from the set of battery cells **122** occupying the battery tray **120**; and a container configured to enclose the battery rack **11** including a perforated floor. The perforated floor **112** is arranged below the battery rack **11**. The system further includes a waste tank **160** arranged below the container; defining a fluid capacity; and configured to receive fluid from the container via the perforated floor.

(149) In one variation wherein a first battery tray **120** including a perforated tray base **126** is arranged in a battery rack **111** above a second battery tray **120**, the system **100** can further include a catch basin, arranged below the perforated tray base **126** of the first battery tray **120**, and

configured to receive fluid from the first battery tray **120**. The catch basin can be further configured to direct fluid into a tube, drain, or onto the perforated floor, away from the second battery tray **120**. In another variation, the system **100** is configured to direct fluid into the first battery tray **120** including the perforated tray base **126** after the first battery tray **120** is ejected from the battery rack **111**. Fluid released from the first battery rack **111** via the perforated tray base **126** drains directly onto the perforated floor **112** of the container and is captured by the waste tank.

(150) Therefore, the system **100** can capture fluid runoff from the container, including fluid released by nozzles **114** arranged within the container, and external fluids such as fire suppressants, introduced into the container from external sources (e.g., a fire suppressant fluid directed into the container by a firefighting team.) The system **100** can capture and isolate fluid that can contain contaminants or chemicals harmful to or undesirable in the external environment within the waste tank. The inner surface of the waste tank **160** can also be configured to resist adverse reactions with fluid or contaminants, thereby enabling a firefighting team to effectively suppress a fire event in the container without limiting application of fire suppressant fluids or chemicals. The waste tank **160** can be further configured to isolate fluid for an extended period of time, and therefore be emptied at a time following suppression of the fire event, when the risk to a human of approaching the container is below a risk threshold.

(151) 11.2 Interconnected Waste Tanks

(152) In one variation as shown in FIG. **5**, in which a first battery storage container **110** is located proximal a second battery storage container **110**, the first waste tank **160** of the first battery storage container **110** can be connected via a pipe to a second waste tank **160** of a second battery storage container **110**. In this configuration, the first waste tank **160** fills with fluid during a fire event, such as when additional fluid is deposited in the battery storage container **110** (i.e., by a firefighting team responding to the fire event). This fluid can then flow from the first waste tank, through the pipe, into the second waste tank. The second waste tank **160** can therefore function as an overflow waste tank **160** for the first battery storage container **110**, thereby preventing fluid from over-topping and spilling out of the first waste tank **160** and contaminating the environment external to the first battery storage container **110**.

(153) For example, the system **100** can include a first waste tank, arranged below a first container, defining a first fluid capacity, and configured to receive fluid from the first container via a perforated floor. The system **100** can further include a second waste tank: arranged external to the first container; defining a second fluid capacity; fluidly coupled to the first waste tank; and configured to receive fluid from the first waste tank **160** in response to the first waste tank **160** receiving a volume of fluid greater than the first fluid capacity.

(154) In another implementation, the second waste tank **160** can be a self-contained waste tank **160** arranged external to the first waste tank, disconnected from a second battery storage container **110**. In another implementation, an array of waste tanks, each fluidly coupled to a battery storage container no, can be fluidly interconnected to share fluid directed into a first battery storage container **110** amongst the array of waste tanks, increasing the volume of fluid that can be introduced into the first battery storage container no during suppression of a fire event.

(155) Therefore, a battery storage container no can be coupled to a first waste tank, the first waste tank **160** further interconnected to a second waste tank, enabling the first battery storage container no to receive a volume of fluid greater than the fluid capacity of the first waste tank **160** without releasing contaminated fluid into the external environment. Additional fluid capacity can enable a response team, such as a firefighting team, to introduce a greater amount of fire suppressants to suppress a fire event in the battery storage container **110** without releasing potential contaminants.

(156) 12. Explosion Mitigation

(157) As shown in FIGS. **2** and **7**, The system **100** can further include a door **115** arranged at the perimeter wall of the housing of the battery storage container **110** and can be configured to open when the controller **130** detects a change in ambient conditions within the storage chamber

indicative of a potential explosive event such as the detection of hazardous gasses (i.e., hydrogen, methane, hydrocarbons, carbon monoxide.)

(158) For example, the system **100** can include the sensor **113**, configured to detect a precursor condition **102** to an incipient explosion event in the container, and the door **115** configured to transition from the closed position to the open position in response to the detection of the precursor condition **102** in the container. The door **115** transitioning to the open position exposes the interior of the container to the exterior environment, thereby venting any gasses in the interior of the container to the exterior atmosphere.

(159) In one implementation, the system can include an actuator (e.g., a spring mechanism, a piston) mounted between the door **115** and the housing. when the door **115** is closed, the actuator is under tension. When the actuator extends, the door **115** is forced by the actuator into the open position.

(160) In another implementation, the system includes an electronically-controllable retention mechanism—such as an electromagnetic latch or an electrical fusible link—mounted between the door **115** and the housing and configured to retain the door **115** in the closed position. The retention mechanism is controllable by the controller **130** and can be released in response to detection of an increase in the concentration of gases within the unit. The door, actuator, and retention mechanism are arranged in a fail-safe configuration, in which a loss of power causes the retention mechanism to release, thereby causing the door **115** to open under the force of the actuator.

(161) Upon release of the retention mechanism, the door **115** opens under the force of the actuator, exposing the interior of the storage chamber to the ambient atmosphere and venting the gases within the storage chamber, thereby preventing an increase in the volume and pressure of flammable gas within the storage chamber that could lead to an explosive event and/or potential rupture or destruction of the storage chamber.

(162) In another implementation wherein the container cannot accommodate a door **115** arranged in the container sidewall, the container can include an operable vent, configured to transition between a closed state, isolating the interior and exterior of the container, and an open state, exposing the interior of the container to the external environment. The vent defines an area proportionate to the volume of the container to provide a volume of airflow through the vent to replace the volume of gas in the container within a minimum time (e.g., less than 60 seconds) to prevent a buildup of volatile gasses within the container during a period of time between a first time when the precursor condition **102** is detected, and a second time when sufficient gas exchange can occur to reduce the concentration of volatile gas below a threshold concentration.

(163) The system **100** can actively detect a potential explosive event within a battery tray **120** by monitoring outputs of multiple sensors including a gas sensor, pressure sensor, humidity sensor, and/or a temperature sensor. The controller **130** can activate the retaining mechanism to release the door **115** in response to a sensor **113** detecting a change in ambient conditions within the battery storage container **110** that precede an explosive event, such as detecting elevated temperatures, gas production, detecting smoke, and/or specific gas constituents.

(164) For example, the system **100** can include: the container configured to enclose the battery rack **111**; a sensor **113** configured to detect presence of a volatile gas indicating an incipient explosion event in the container; and the door **115** arranged in the container opening in the sidewall of the container. The door **115** is configured to: seal the container opening, isolating the interior of the container from the exterior of the container in the closed position; expose the container interior to the container exterior in the open position; and transition from the closed position to the open position in response to detection by the sensor **113** of the presence of the gas indicating an incipient explosion event in the container.

(165) In this example, the system is configured to detect the presence of a volatile gas in the battery storage container **110** that is not present during nominal operating conditions of the set of battery cells **122**. The sensor **113** can be configured to detect a particular volatile gas indicating an

incipient explosion event based on the characteristics of the particular battery cells **122** in the container. In response to detection of the presence of the particular volatile gas, the controller **130** can detect a precursor condition **102** to an incipient explosion and trigger the door **115** to open, thereby venting the volatile gas to the exterior of the battery storage container **110**.

(166) In another variation in which the system **100** detects presence of a flammable gas in the container, in addition to triggering the door **115** to transition from the closed position to the open position, the controller **130** is further configured to cut power to the battery storage container **110** to reduce the possibility of electrical shorts in the container with the potential to generate a spark capable of igniting the flammable gas and causing an explosion in the container.

(167) In another variation of this example, the system **100** includes a sensor **113** configured to detect a volatile gas concentration within the container greater than a threshold gas concentration. The controller **130** is configured to: receive a signal from the sensor **113**; detect the signal as a precursor to an explosion event; and, in response to detecting the precursor to the explosion event, trigger the door **115** to transition from the closed position to the open position.

(168) In this implementation, the system **100** is configured to detect an increased concentration of a particular volatile gas or a particular set of volatile gasses that can be present in the container during nominal operating conditions in trace amounts and/or in concentrations below an operating threshold concentration. The system **100** is configured to detect an increase in concentration of the particular volatile gas beyond the operating threshold concentration and, in response, initiate a mitigation action (such as triggering the door **115** to transition to the open position) to reduce the concentration of the volatile gas prior to the occurrence of an explosion event. In one variation of this implementation, the system **100** can be configured to detect a concentration of a first volatile gas, nominally non-reactive in the battery storage container **110**, and a concentration of a second volatile gas reactive with the first volatile gas, the concentration of the first volatile gas and the concentration of the second volatile gas representing a precursor condition **102** to an explosion event.

(169) After the door **115** has been opened, the controller **130** can continue to monitor the ambient conditions within the battery storage container **110**. More specifically, the controller **130** can monitor the gas concentration, pressure, and constituents within the storage chamber and record these data in a remote database. In one variation in which the controller **130** is configured to automatically alert and/or dispatch a fire department to the location of the battery storage container **110**, the controller **130** can transmit the data to the emergency responders en route to the battery storage container **110**.

(170) For example, the controller **130** can release the door **115** in response to the detection of flammable and/or explosive gasses within the battery storage container **110**, which can be products of battery cell **122** failure such as hydrogen, methane, or hydrocarbons. By releasing the door **115** and venting these gasses to the ambient atmosphere upon detection, the system prevents the buildup of flammable/explosive gasses within the confined environment of the battery storage container **110**, thereby reducing the possibility of ignition of concentrated flammable/explosive gasses in a confined space and possible explosion. Further, the controller **130** releases the door **115** in response to gas concentration below the lower flammability limit to prevent a dangerous buildup of flammable/explosive gasses. Generally, firefighters will not approach a scene if the concentration of gasses were to exceed the lower flammability limit, as it presents too great a hazard to the firefighters. Therefore, by releasing the door **115** upon detection of flammable/explosive gasses and venting them to the ambient atmosphere, the system prevents a situation in which firefighters cannot approach a scene to mitigate the fire event.

(171) In a related example, the system **100** is configured to open the door **115** to release gasses within the container to the atmosphere and eject a battery tray **120** exhibiting a precursor condition **102** or a fire event. In this example, the system **100** can include: a battery rack **111** configured to support a first battery tray **120** and a second battery tray **120**; the container configured to enclose

the battery rack **111**; and the door, arranged in an opening in the container sidewall, and configured to transition from the closed position to the open position. In response to a precursor condition **102** detected in the first battery tray **120** by the sensor **113**, the door **115** transitions from the closed position to the open position, and the first battery tray **120** is extended from the battery rack **111** through the container opening. Extending the battery tray **120** through the opening in the container increases the distance between the set of battery cells **122** in the first battery tray **120** and a second set of battery cells **122** in the second battery tray **120**. Additionally, extending the first battery tray **120** through the opening in the battery storage container no enables a firefighting team to approach and suppress the fire event from additional sides.
(172) In one variation, the controller **130** can activate the retaining mechanism to release the door **115** in response to a sensor **113** detecting a change in ambient conditions within the battery storage container no that precede an explosive event, such as detecting an increase in the temperature within the storage chamber beyond a threshold (i.e., beyond 100 degrees Celsius)
(173) In another variation, the door **115** is mounted within the perimeter wall of the housing such that the gravitational force on the door **115** forces the door **115** into the open position and the retention mechanism holds the door **115** in the closed position. In another variation, a weight is attached to the door **115** mechanism such that the gravitational force on the weight forces the door **115** into the open position and the retention mechanism holds the door **115** in the closed position.
(174) Therefore, the system **100** can detect various precursor conditions **102** indicative of an incipient or imminent explosion event in the battery storage container **110** and initiate mitigation actions to slow or prevent progression to an explosion within the battery storage container **110**. Generally, the system is configured to detect a precursor condition **102** and execute a mitigation action to suppress the precursor condition **102** before progression to an explosion event. Additionally, the system can be configured to execute mitigation actions in a sequence of increasing potency and/or destructiveness to increase the probability of preventing progression to the explosion event, while protecting a maximum number of elements of the system **100** not exhibiting a precursor condition **102** from secondary damage caused by the mitigation action.
(175) 14. Additional Applications
(176) In yet another example, the system **100** can be integrated within a battery and/or hybrid electric vehicle, which use lithium battery cells **122** to power an electric motor of the vehicle. In this example, the electric vehicle can include: a cooling channel **151** that is configured to receive a fire suppression fluid (e.g., water vapor); and a nozzle **114** positioned at the distal end of the cooling channel **151**, the nozzle **114** including an inlet connected to the cooling channel **151** and an outlet positioned to direct a fluid spray pattern toward a battery of the vehicle; and a controller **130** configured to detect the presence of a fire event of the battery within the vehicle. The controller **130** can further be configured to transmit a warning prompt to a dashboard display of the vehicle to notify a driver of the vehicle and/or a remote monitoring system in order to notify a manufacturer of the vehicle and/or a fire department service. In one variation of this example implementation, the vehicle can include an onboard fluid storage tank in order to store and distribute the fire suppression fluid. Alternatively, the vehicle can include a fluid inlet that receives and directs externally provided fluid (e.g., water from a fire truck) into the cooling channel **151**, through the nozzle **114**, and to the battery cells **122**.
(177) In yet another example implementation, the system **100** can be configured to affix to a system with an existing cooling and fluid distribution apparatus (e.g., HVAC system). In this example, the system **100** can include: a cooling channel **151**; and a nozzle **114** positioned to direct a fluid spray pattern at a power source of the system within the existing cooling system. Thus, the system **100** can integrate into the structure (e.g., power, sensor **113***s*, distribution) of the existing cooling apparatus and provide additional fire protection for assets serviced by the existing cooling apparatus.
(178) The systems and methods described herein can be embodied and/or implemented at least in

part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions can be executed by computer-executable components integrated with the application, applet, host, server, network, website, communication service, communication interface, hardware/firmware/software elements of an owner computer or mobile device, wristband, smartphone, or any suitable combination thereof. Other systems and methods of the embodiment can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions can be executed by computer-executable components integrated by computer-executable components integrated with apparatuses and networks of the type described above. The computer-readable medium can be stored on any suitable computer readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, or any suitable device. The computer-executable component can be a processor but any suitable dedicated hardware device can (alternatively or additionally) execute the instructions.
(179) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

Claims

1. A system for detecting and mitigating a fire within a battery storage container comprising: a first battery rack; a first sensor configured to detect a precursor condition to an incipient fire event in the first battery rack; a first battery tray: configured to retain a first set of battery cells; occupying the first battery rack in an inserted position; and extending out of and supported by the first battery rack in an extended position; a first tray ejector configured to transition the first battery tray from the inserted position to the extended position in response to detection of the precursor condition in the first battery rack; a first intercooler: arranged in the first battery tray; and comprising a first cooling channel configured to circulate a fluid to cool the first set of battery cells occupying the first battery tray; a supply manifold arranged proximal the first battery rack, fluidly coupled to the first intercooler, and configured to supply the fluid to the first intercooler; a return manifold arranged proximal the first battery rack, fluidly coupled to the first intercooler, and configured to receive the fluid from the first intercooler; and a first nozzle: fluidly coupled to the supply manifold; arranged in the first battery tray; and configured to receive the fluid from the supply manifold and direct the fluid into the first battery tray in response to detection of the precursor condition in the first battery rack.
2. The system of claim 1: wherein the first sensor is arranged in the first battery rack adjacent the first battery tray; and further comprising: a second battery tray: configured to retain a second set of battery cells; occupying the first battery rack in an inserted position; and extending out of and supported by the first battery rack in an extended position; a second sensor arranged in the battery rack adjacent the second battery tray and configured to detect the precursor condition; a second tray ejector configured to transition the second battery tray from the inserted position to the extended position in response to detection of the precursor condition; a second intercooler: arranged in the second battery tray; and comprising a second cooling channel configured to circulate the fluid to cool the second set of battery cells occupying the second battery tray; a second nozzle: fluidly coupled to the supply manifold; arranged in the second battery tray; and configured to receive the fluid from the supply manifold and direct the fluid into the second battery tray in response to detection of the precursor condition; and a controller configured to: receive a first signal from the first sensor and a second signal from the second sensor; detect the precursor condition in the first battery tray based on the first signal; and in response to detecting the precursor condition in the first battery tray: trigger the first tray ejector to transition the first battery tray from the inserted position to the extended position; and trigger the first nozzle to direct the fluid into the first battery tray to

suppress the precursor condition in the first battery tray.
3. The system of claim 2: wherein the first sensor is configured to: detect a temperature of a first battery cell, in the first set of battery cells, occupying the first battery tray; and transmit the first signal to the controller; and wherein the controller is configured to: receive the first signal from the first sensor; detect the temperature of the first battery cell exceeding a threshold temperature based on the first signal; and in response to detecting the temperature of the first battery cell exceeding the threshold temperature in the first battery tray: trigger the first tray ejector to transition the first battery tray from the inserted position to the extended position; and trigger the first nozzle to direct the fluid into the first battery tray to suppress the precursor condition in the first battery tray.
4. The system of claim 2 wherein: the first sensor is configured to detect a pressure of a first battery cell, in the first set of battery cells, occupying the first battery tray; and the controller is configured to: receive the first signal from the first sensor; detect the pressure of the first battery cell exceeding a threshold pressure based on the first signal; and in response to detecting the pressure of the first battery cell exceeding the threshold pressure in the first battery tray: trigger the first tray ejector to transition the first battery tray from the inserted position to the extended position; and trigger the first nozzle to direct the fluid into the first battery tray to suppress the precursor condition in the first battery tray.
5. The system of claim 2: further comprising: a third battery tray: configured to retain a third set of battery cells; occupying a second battery rack in an inserted position; and extending out of and supported by the second battery rack in an extended position; a third sensor: arranged external to the first battery rack and the second battery rack; and configured to detect a temperature within the third battery tray; and a third tray ejector configured to transition the third battery tray from the inserted position to the extended position; and wherein the controller is further configured to: receive a third signal from the third sensor; detect the precursor condition in the third battery tray based on the third signal; and in response to detecting the precursor condition in the third battery tray, trigger the third tray ejector to transition the third battery tray from the inserted position to the extended position.
6. The system of claim 1: further comprising: a power bus arranged in the first battery rack adjacent the first battery tray; and an electrical disconnect: electrically coupled to and interposed between the power bus and the first set of battery cells within the first battery tray, and configured to: in a coupled state, electrically couple the power bus to the first set of battery cells; in a decoupled state, electrically disconnect the power bus from the first set of battery cells; and transition from the coupled state to the decoupled state; and a controller configured to trigger the electrical disconnect to transition from the coupled state to the decoupled state in response to detection of the precursor condition in the first battery tray.
7. The system of claim 1: further comprising: a container configured to enclose the first battery rack; and a door: arranged in a container opening in a sidewall of the container; and configured to: seal the container opening, isolating an interior of the container from an exterior of the container in a closed position; expose the container interior to the container exterior in an open position; and transition from the closed position to the open position; and wherein the first battery tray is configured to extend out of the first battery rack through the container opening.
8. The system of claim 7, wherein: the first sensor is configured to detect a precursor condition to an incipient explosion event in the container; and the door is configured to transition from the closed position to the open position in response to the detection of the precursor condition to an incipient explosion event in the container.
9. The system of claim 8: wherein the first sensor is configured to detect a volatile gas concentration within the container; and further comprising a controller configured to: receive a signal from the first sensor; detect the signal as a precursor to an explosion event; and in response to detecting the precursor to the explosion event, trigger the door to transition from the closed position to the open position.

10. The system of claim 1: wherein the first cooling channel further defines a set of apertures configured to release fluid from the first cooling channel into the first battery tray; and further comprising a set of meltable plugs, each meltable plug in the set of meltable plugs configured to: insert into an aperture in the first cooling channel; seal the aperture when a temperature of the first battery tray is maintained below a threshold temperature, retaining fluid within the first cooling channel; and melt in response to the temperature in the first battery tray exceeding the threshold temperature, releasing the fluid into the first battery tray.
11. The system of claim 1: wherein the first cooling channel further defines a set of apertures configured to release fluid from the first cooling channel into the first battery tray; and further comprising: a set of pressure sensitive plugs, each pressure sensitive plug in the set of pressure sensitive plugs configured to: insert into an aperture in the first cooling channel; seal the aperture when a pressure of fluid in the first cooling channel is maintained below a threshold pressure, retaining fluid in the first cooling channel; and eject from the aperture in response to the pressure of fluid in the first cooling channel increasing to greater than the threshold pressure, releasing fluid into the first battery tray; and a pump: fluidly coupled to the first intercooler; and configured to increase the pressure of fluid in the first cooling channel above the threshold pressure in response to detection of the precursor condition by the first sensor.
12. The system of claim 1: further comprising a directional control valve: fluidly coupled to and interposed between: the supply manifold and the first intercooler; and the supply manifold and the first nozzle; and configured to: in a first state, receive fluid from the supply manifold and direct fluid to the first intercooler; and in a second state, receive fluid from the supply manifold and direct fluid to the first nozzle; and wherein triggering the first nozzle to direct fluid into the first battery tray comprises triggering the directional control valve to transition from the first state to the second state.
13. The system of claim 1: further comprising a second nozzle: arranged above the first battery rack and laterally offset from the first battery rack; and configured to direct fluid into the first battery tray when the first battery tray is in an ejected position; and wherein: at a first time, the first sensor is configured to detect the precursor condition to the incipient fire event; at a second time following the first time, the first battery tray is configured to eject from the first battery rack; and at a third time following the second time, the second nozzle is configured to release fluid into the first battery tray.
14. The system of claim 13: wherein the first battery tray further defines a perforated base configured to release fluid from the first battery tray; and further comprising: a first container: configured to enclose the first battery rack; and comprising a perforated floor: arranged below the first battery rack; and configured to release fluid through the floor; a first waste tank: arranged below the first container; defining a first fluid capacity; and configured to receive fluid from the first container via the perforated floor; and a second waste tank: arranged external to the first container; defining a second fluid capacity; fluidly coupled to the first waste tank; and configured to receive fluid from the first waste tank in response to the first waste tank receiving a volume of fluid greater than the first fluid capacity.
15. A system for detecting and mitigating a fire within a battery storage container comprising: at a battery tray arranged within a battery rack: retaining a set of battery cells within the battery tray; and circulating a fluid through an intercooler arranged within the battery tray to cool the set of battery cells occupying the battery tray; at a directional control valve fluidly coupled to a supply manifold, the intercooler and a nozzle arranged within the battery tray: receiving fluid from the supply manifold; and supplying fluid to the intercooler; at a sensor arranged within the battery tray, detecting a precursor condition to an incipient fire event in the first battery tray; in response to detection of the precursor condition by the sensor, extending the battery tray from the battery rack; in response to detection of the precursor condition by the sensor, controlling the directional control valve to transition from a first state supplying fluid to the intercooler, to a second state supplying

the fluid to the nozzle; and at a controller: receiving a signal from the sensor indicating detection of the precursor condition to an incipient fire event by the sensor; in response to receiving the signal from the sensor, generating a notification indicating the precursor condition present in the battery tray; and transmitting the notification to an operator.

16. A system for detecting and mitigating a fire within a battery storage container comprising: a first battery rack; a first battery tray: configured to retain a first set of battery cells; occupying the first battery rack in an inserted position; and extending out of and supported by the first battery rack in an extended position; a first sensor configured to: at a first time preceding ejection of the first battery tray from the first battery rack, detect a first temperature within the first battery tray below a threshold temperature; at a second time following the first time, detect a second temperature within the first battery tray exceeding the threshold temperature; and at a third time following the second time and following ejection of the first battery tray from the first battery rack, detect a third temperature within the first battery tray falling below the threshold temperature; and a controller configured to: at a fourth time following the second time and preceding the third time: receive a first signal, from the first sensor, following detection of the second temperature in the first battery tray; and detect a first precursor condition in the first battery tray based on the first signal; at a fifth time following the third time, receive a second signal from the first sensor; calculate a risk value of a human approaching the first battery tray based on the second signal, a composition of the first set of battery cells, and a difference between the fifth time and a present time; detect the risk value falling below a threshold risk value; and in response to detecting the risk value falling below the threshold risk value: generate a prompt to approach the first battery tray and replace the first set of battery cells in the first battery tray, including the risk value; and transmit the prompt to an operator; and a first tray ejector configured to transition the first battery tray from the inserted position to the extended position in response to detection of the first precursor condition in the first battery tray.

17. The system of claim 16: further comprising a container configured to enclose the first battery rack; wherein the controller is configured to detect presence of a volatile gas indicating an incipient explosion event in the container; and further comprising a door: arranged in a container opening in a sidewall of the container; and configured to: seal the container opening, isolating an interior of the container from an exterior of the container in a closed position; expose the container interior to the container exterior in an open position; and transition from the closed position to the open position in response to detection of the presence of the gas indicating an incipient explosion event in the container by the controller.

18. The system of claim 16: further comprising: a first nozzle: arranged within the first battery tray; and configured to direct fluid into the first battery tray; a second battery tray: arranged adjacent to the first battery tray; configured to retain a second set of battery cells; occupying the first battery rack in an inserted position; and extending out of and supported by the first battery rack in an extended position; a second sensor; a second tray ejector configured to transition the second battery tray from the inserted position to the extended position; a second nozzle: arranged within the second battery tray; and configured to direct fluid into the second battery tray; a third battery tray: arranged adjacent to the second battery tray; configured to retain a third set of battery cells; occupying the first battery rack in an inserted position; and extending out of and supported by the first battery rack in an extended position; and a third tray ejector configured to transition the third battery tray from the inserted position to the extended position; and wherein the controller is configured to: receive a first signal from the first sensor at the first time; detect the first precursor condition to a first incipient fire event in the first battery tray at the fourth time based on the first signal; in response to detecting the first precursor condition in the first battery tray: trigger the first tray ejector to transition the first battery tray from the inserted position to the extended position; and trigger the first nozzle to direct the fluid into the first battery tray to suppress the first precursor condition in the first battery tray; receive a second signal from the second sensor at a sixth time

following the fourth time; detect a second precursor condition to a second incipient fire event in the second battery tray at a seventh time following the sixth time based on the second signal; and in response to detecting the second precursor condition in the second battery tray: trigger the third tray ejector to transition the third battery tray from the inserted position to the extended position; and trigger the second nozzle to direct the fluid into the second battery tray to suppress the second precursor condition in the second battery tray.

19. The system of claim 16, further comprising: a power bus arranged proximal the first battery rack; and an electrical disconnect: electrically coupled to and interposed between the power bus and the first set of battery cells within the first battery tray; and configured to: in a coupled state, electrically couple the power bus to the first set of battery cells; in a decoupled state, physically and electrically disconnect the power bus from the first set of battery cells; and transition from the coupled state to the decoupled state in response to ejection of the first battery tray from the first battery rack.